VIGNAN'S
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(22BI401) NEXT GENERATION SEQUENCING

MODULE- 1

T5- PRACTICES

DEPARTMENT OF BIOINFORMATICS

SUBMITTED BY: -

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PRACTICE: -1

RETRIEVAL OF GENOMIC DATA USING SRA TOOLKIT

OBJECTIVE: -

The primary objective of this task is to efficiently retrieve genomic data from the Sequence Read Archive (SRA) using the SRA Toolkit. The retrieved data will be in FASTQ format, which is a standard format for storing high-throughput sequencing data, and can be further processed or analyzed for downstream bioinformatics applications. Specifically, the goals are:

1. **Installing SRA Toolkit:** Ensure that the necessary software for retrieving and processing SRA data is available and correctly configured on the local system.

2. **Configure the SRA Toolkit:** Set up the toolkit to ensure it can access the SRA database and handle the data efficiently.

3. **Download Genomic Data:** Use the `fastq-dump` command to download the genomic data. This involves converting the data from SRA format to FASTQ format, which is more widely used and compatible with various bioinformatics tools.

4. **Optimize Data Retrieval:** Implement options to compress the data (using gzip) and handle paired-end data if necessary, ensuring that the downloaded data is in the desired format and ready for analysis.

5. **Documentation and Automation:** Provide clear instructions and a script to automate the download process, ensuring reproducibility and ease of use for future data retrieval tasks.

By achieving these goals, researchers will have access to high-quality genomic data in a format suitable for further analysis, enabling them to proceed with their specific research objectives.

SOFTWARE USED:

Sratoolkit(https://github.com/ncbi/sra-tools/wiki/01.-Downloading-SRA-Toolkit)

1. Navigate to your project folder

cd C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS

- This makes sure all work happens in your project directory.

2. Create a custom folder for SRA data

mkdir keer_sra_data

- Makes a dedicated folder named `keer_sra_data` where your `.sra` and `.fastq` files will be stored.

3. Go to the bin folder of SRA Toolkit

cd C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin

- The bin folder contains all executable tools (prefetch, fasterq-dump, etc.). All main commands are run from here.

4. Download the .sra file using Prefetch

prefetch SRR957824 -O ..\..\..\keer_sra_data

- What happens here:

- prefetch → tool to download SRA files.
- SRR957824 → accession number (unique ID for your dataset).
- -O ..\..\..\keer_sra_data → tells the tool to save the downloaded data in your custom folder keer_sra_data.

After this, the .sra file will be saved in:

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data\SRR957824\SRR957824.sra

1. **C:\Users\keert>cd C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS**
2. **C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS>mkdir keer_sra_data**
3. **C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS>cd sratoolkit.3.2.1-win64**
4. **C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64>cd bin**
5. **C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>prefetch SRR957824 -O ..\..\..\keer_sra_data**
6. **C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data\SRR957824\SRR957824.sra**

```
C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64>KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin
The system cannot find the path specified.

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64>cd sratoolkit.3.2.1-win64

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64>dir
 Volume in drive C is Windows
 Volume Serial Number is 90F5-4C94

 Directory of C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64

18-09-2025  13:00    <DIR>          .
18-09-2025  13:00    <DIR>          ..
18-09-2025  13:00    <DIR>          bin
18-09-2025  13:00            37,144 CHANGES.txt
18-09-2025  13:00    <DIR>          example
18-09-2025  13:00             1,711 README-blastn.txt
18-09-2025  13:00             5,890 README-vdb-config.txt
18-09-2025  13:00            11,696 README.md
18-09-2025  13:00    <DIR>          schema
               4 File(s)         55,641 bytes
               5 Dir(s)  29,031,596,032 bytes free

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64>cd bin

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>prefetch SRR957824 -O ..\..\..\keer_sra_data
2025-09-18T07:47:33 prefetch.3.2.1: 1) Resolving 'SRR957824'...
2025-09-18T07:47:38 prefetch.3.2.1: Current preference is set to retrieve SRA Normalized Format files with full base quality scores
2025-09-18T07:47:40 prefetch.3.2.1: 1) Downloading 'SRR957824'...
2025-09-18T07:47:40 prefetch.3.2.1:  SRA Normalized Format file is being retrieved
2025-09-18T07:47:40 prefetch.3.2.1:  Downloading via HTTPS...
2025-09-18T07:56:18 prefetch.3.2.1:  HTTPS download succeed
2025-09-18T07:56:18 prefetch.3.2.1:   verifying 'SRR957824'...
2025-09-18T07:56:20 prefetch.3.2.1:  'SRR957824' is valid: 296295241 bytes were streamed from 296288633
2025-09-18T07:56:20 prefetch.3.2.1: 1) 'SRR957824' was downloaded successfully
2025-09-18T07:56:20 prefetch.3.2.1: 1) Resolving 'SRR957824's dependencies...
2025-09-18T07:56:20 prefetch.3.2.1: 'SRR957824' has 0 unresolved dependencies

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>
```

PRACTICE-2

PRE PROCESSING AND QULAITY ASSESSMENT BY USING FASTQC

OBJECTIVE

The primary objective of performing a quality check with FastQC is to assess the overall quality of the raw sequencing data obtained from high-throughput sequencing experiments. This involves generating comprehensive quality control (QC) reports that highlight various metrics and potential issues within the data. The specific goals of this quality check are:

1. **Identify Potential Issues:** Detect common issues such as low-quality bases, adapter contamination, overrepresented sequences, and sequence duplication levels that could affect downstream analyses and results.

2. **Assess Data Quality:** Evaluate the overall quality of the sequencing data by examining various quality metrics, including per base sequence quality, per sequence quality scores, GC content, and sequence length distribution.

3. **Guide Data Processing:** Use the insights gained from the quality reports to inform decisions on data preprocessing steps, such as trimming low-quality bases, removing adapters, and filtering out poor-quality reads.

4. **Ensure Data Integrity:** Confirm that the sequencing data meets the required standards for quality and integrity, ensuring that it is suitable for downstream bioinformatics analyses, such as alignment, variant calling, and transcriptome assembly.

5. **Generate Comprehensive Reports:** Produce detailed and easy-to-interpret QC reports that can be shared with collaborators and included in project documentation, providing a clear overview of the data quality and any necessary corrective actions.

By achieving these goals, researchers can ensure the reliability and accuracy of their sequencing data, leading to more robust and trustworthy scientific conclusions.

SOFTWARE USED:

Fastqc(https://www.bioinformatics.babraham.ac.uk/projects/fastqc/)

PROTOCOL:

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>fastq-dump --stdout -X 2 SRR957824

1. fastq-dump

- `fastq-dump` is a command-line utility that is part of the SRA Toolkit. It is used to convert SRA (Sequence Read Archive) files into FASTQ format, which is a common format for storing sequencing data.

2. --stdout

- The `--stdout` option tells `fastq-dump` to write the output directly to the standard output (the terminal) instead of writing it to a file. This allows you to see the output data in the terminal.

3.-X 2

- The `-X` option is used to limit the number of spots (reads) to dump. In this case, `-X 2` means that only the first 2 spots (reads) will be processed and displayed. This is useful for quickly inspecting the data without processing the entire file.

5. SRR957824

- `SRR957824` is the accession number of the SRA dataset to be processed. This specifies which dataset the `fastq-dump` utility should work on.

Purpose of the Command:

The purpose of this command is to quickly preview the first two reads of the SRA dataset `SRR957824` in FASTQ format. By limiting the output to just 2 reads and displaying them in the terminal, you can get a quick look at the structure and quality of the data without having to process the entire dataset.

```
C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>fastq-dump --stdout -X 2 SRR957824
Read 2 spots for SRR957824
Written 2 spots for SRR957824
@SRR957824.1 1 length=300
ACTATGAGCGAAACGGCATCCTGGCAGCCGAGCGCATCCATTCCTAACTTATTAAAACGCGCGGCGATTATGGCGGAGAACCCTCGTTTCTTTTCCCATCGTGGAGTGCTGGAGGTGGAGACGCCCTGTATGTGCCAGGCGACGGTAACCAAACGT
GTCTCAAACGGGACCAAATGAATATCGGTTACCGTCGCCTGGCTCATACAAGGCGTCTCCACCTCCAGCACTCCACGATCGGCAAAGAAACGACGGATCTCCGCCATAATCGCCGCGCGTTTTACTAAGTTAGGAATGGATGCG
+SRR957824.1 1 length=300
?????BBBDDDDDD8DFFFFFFFFFHHHICEEEEHBHHIIIIIIIFGHHIIIIHFGHHCEEHEEHBEDFEFEFD>B@##########################################################################<????B
?BBBBBBBBEEEECEHF>CFHHHHGHHBHHHHHEHHHHHHHHHHGHHBGHDGHHHHHHHHHHHHHHFHHHHHHHHHHCFHHDEHEDDFFDF@DE=@EBBCEEAEE6?CECE)?E;?????28;?:*:::**0:*0?*:A::AEE#
@SRR957824.2 2 length=300
GATCGGAAGAGCACACGTCTGAACTCCAGTCACAGCGCTATCTCGTATGCCGTCTTCTGCTTGAAAAAAAAAAAAAAAAAAAATCATACTAATATGAGTTCGATATATAAACTGAAAGATCAAACATGAGTTCATCAGTGTACTATATAATCGGG
AGGGGGGCGGGGTGGGGAAGAGGGGAGAATTCGGGGGGGGGCGGGTTATTTAAAAAAAAAAAAAAAAAAAAGAAAATAGAACGTAAAACAAGATGAGATAGACTAAAGTAATCAAGGCTGAGCATATTCTATCTTAGTATAGTT
+SRR957824.2 2 length=300
????????DDDDDDDDGGGGGGIIIIIIHIIHIIHHHHHHHHHIFHHIIHHHHHHIHGHHIIIFGHDHHEHHE#######################################################################
##########################################################################################################################################
```

This allows you to quickly verify that the data is being processed correctly and to inspect the quality and format of the reads.

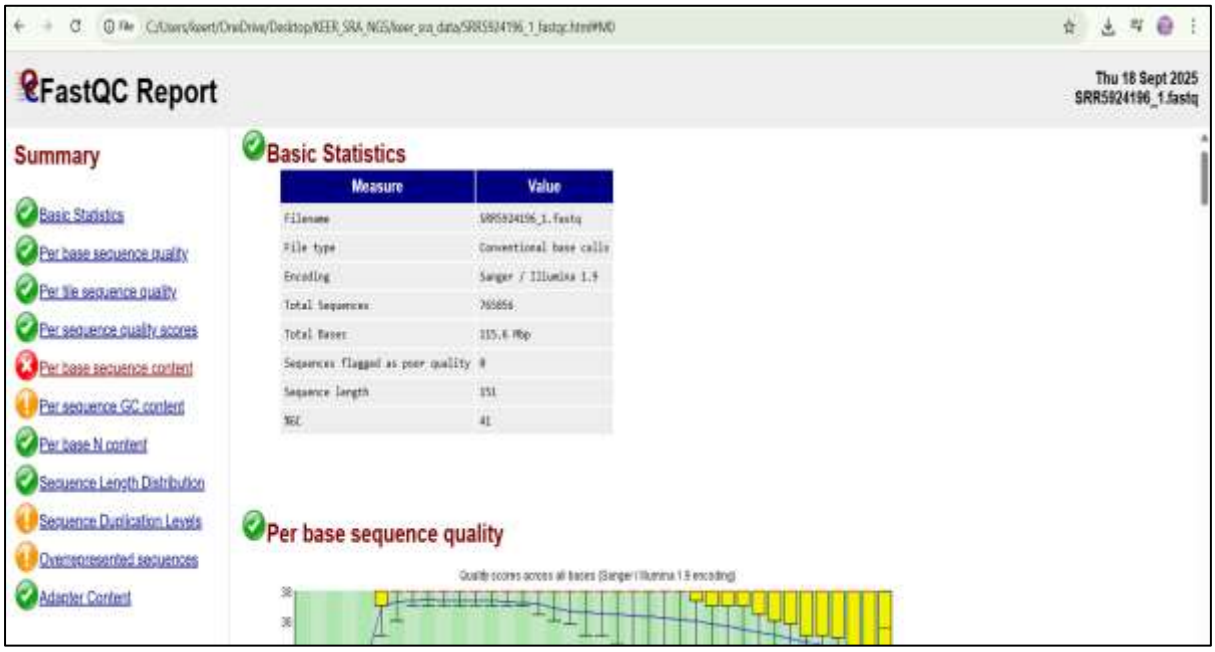
Convert the SRA File to FASTQ Format Using fasterq-dump :-

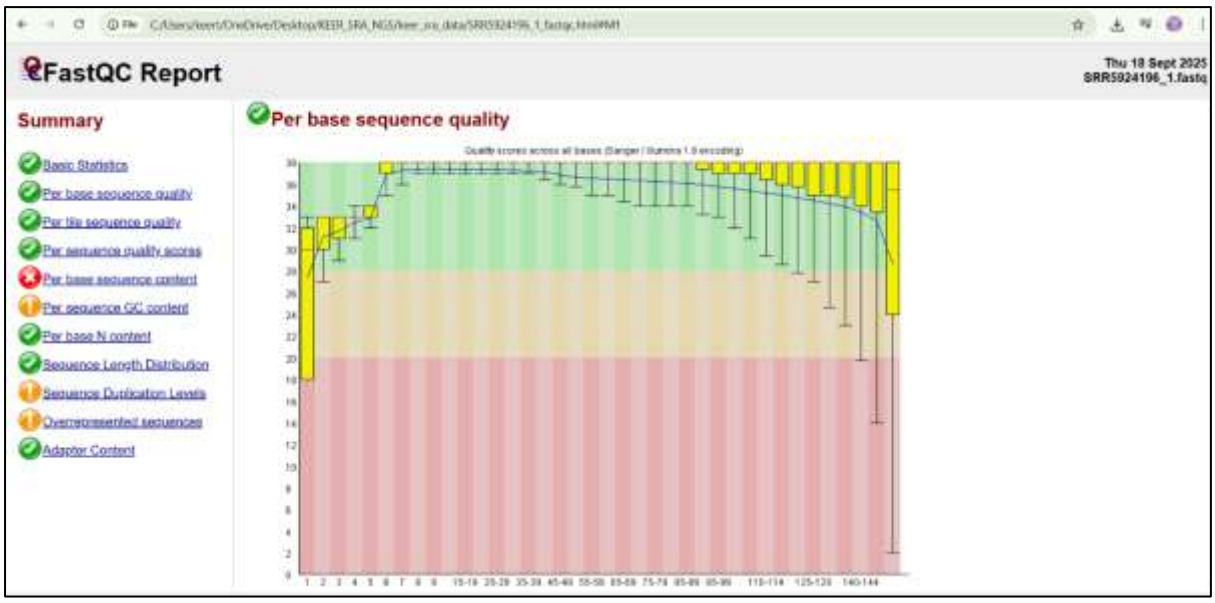
```
C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>fasterq-dump C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\kee
r_sra_data\SRR5924196\SRR5924196.sra -O C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data --split-files --threads 6 --progress
join   :|-------------------------------------------------- 100%
concat :|-------------------------------------------------- 100%
spots read      : 765,856
reads read      : 1,531,712
reads written   : 1,531,712
2025-09-18T10:06:37 fasterq-dump.3.2.1 err: delete_dirs().KDirectoryRemove( /c/Users/keert/OneDrive/Desktop/KEER_SRA_NGS/sratoolkit.3.2.1-win64/sratoolkit.3
.2.1-win64/bin/fasterq.tmp..0/ ) -> RC(rcFS,rcDirectory,rcRemoving,rcDirectory,rcUnauthorized)
2025-09-18T10:06:37 fasterq-dump.3.2.1 err: locked_file_list_delete_dirs().delete_dirs() -> RC(rcFS,rcDirectory,rcRemoving,rcDirectory,rcUnauthorized)

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>fasterq-dump C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\kee
r_sra_data\SRR957824\SRR957824.sra -O C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data --split-files --threads 6 --progress
join   :|-------------------------------------------------- 100%
concat :|-------------------------------------------------- 100%
spots read      : 1,792,335
reads read      : 3,584,670
reads written   : 3,584,670
2025-09-18T10:07:00 fasterq-dump.3.2.1 err: delete_dirs().KDirectoryRemove( /c/Users/keert/OneDrive/Desktop/KEER_SRA_NGS/sratoolkit.3.2.1-win64/sratoolkit.3
.2.1-win64/bin/fasterq.tmp..0/ ) -> RC(rcFS,rcDirectory,rcRemoving,rcDirectory,rcUnauthorized)
2025-09-18T10:07:00 fasterq-dump.3.2.1 err: locked_file_list_delete_dirs().delete_dirs() -> RC(rcFS,rcDirectory,rcRemoving,rcDirectory,rcUnauthorized)

C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\sratoolkit.3.2.1-win64\sratoolkit.3.2.1-win64\bin>
```

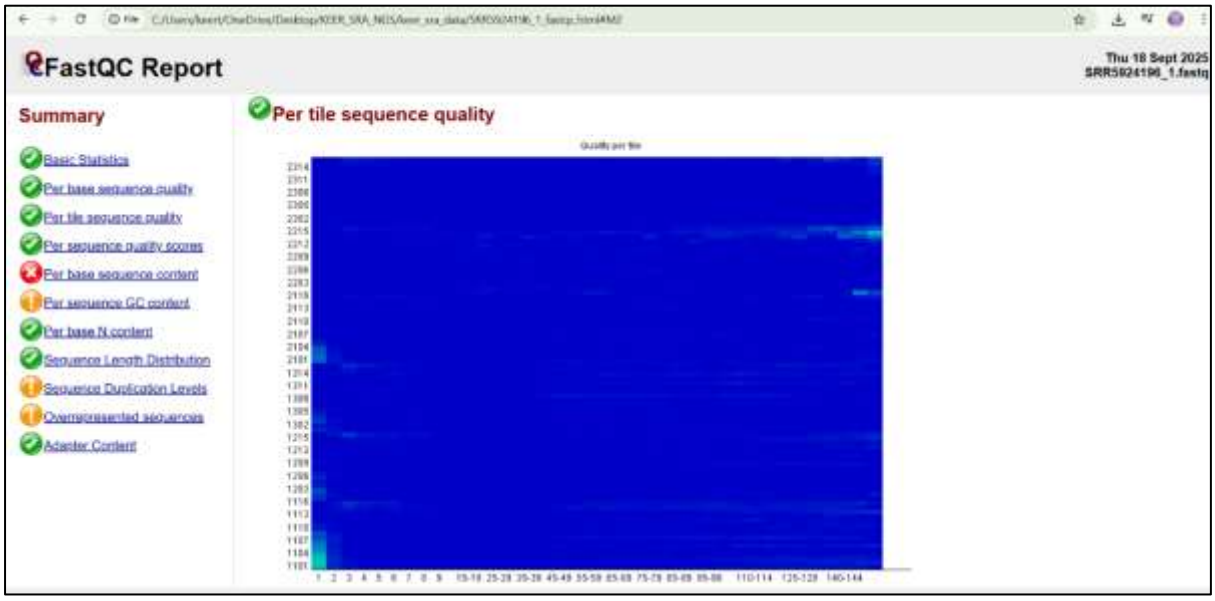
Convert. sra to FASTQ manually

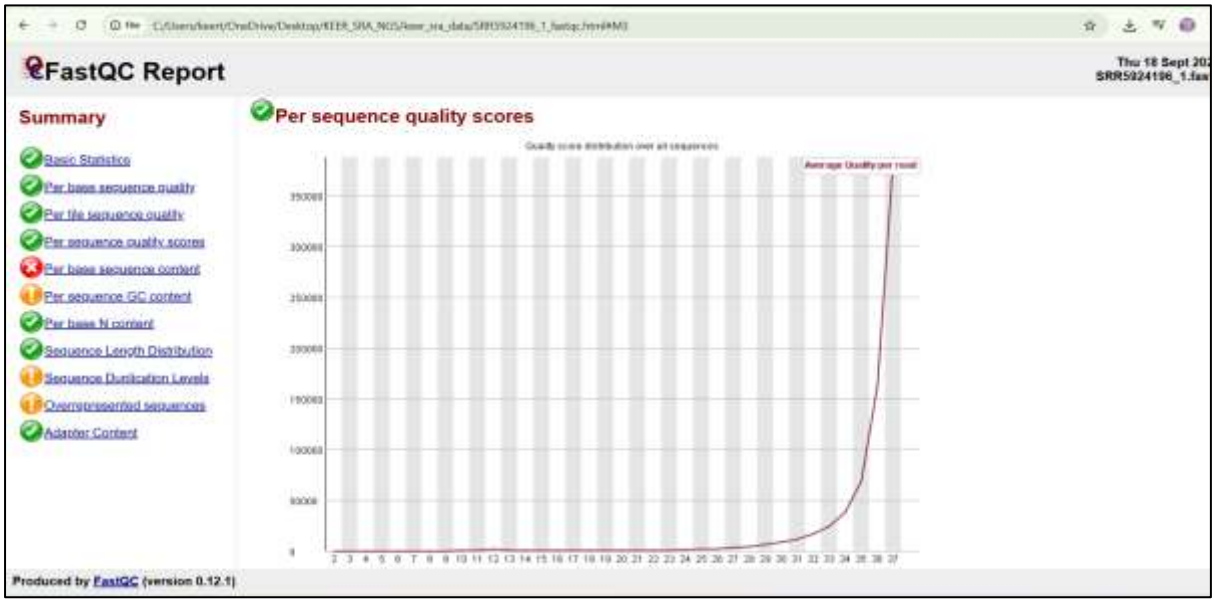
Instead of giving the **accession ID**, give the **full path to the .sra file**:

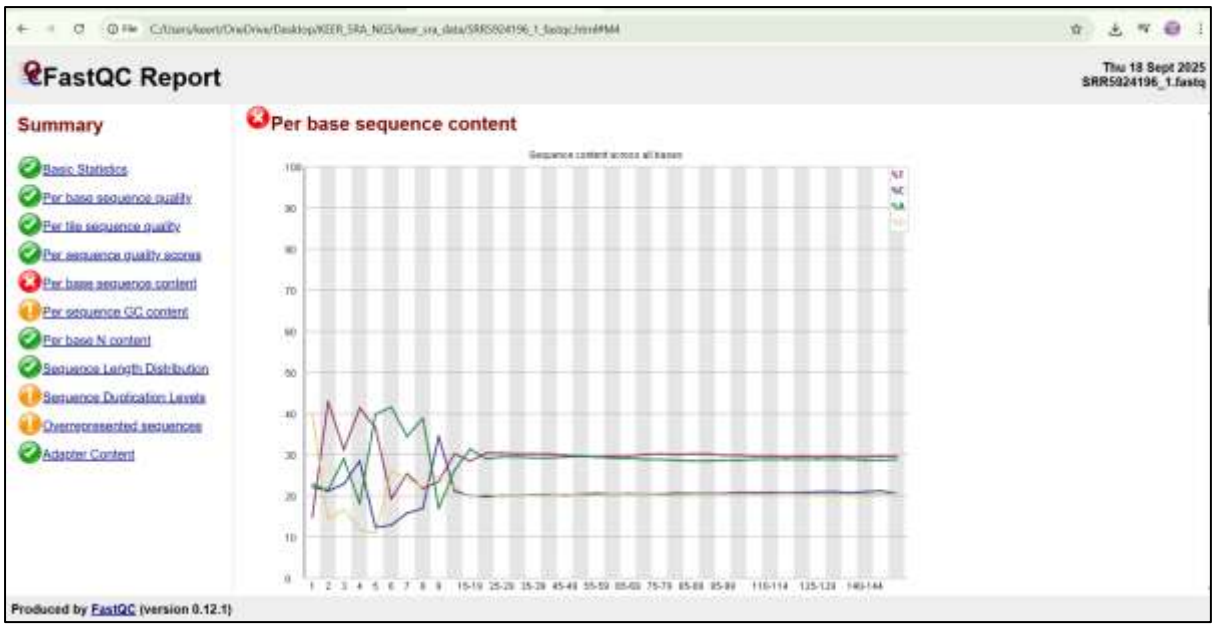
fasterq-dump C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data\SRR5924196\SRR5924196.sra -O C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data --split-files --threads 6 –progress

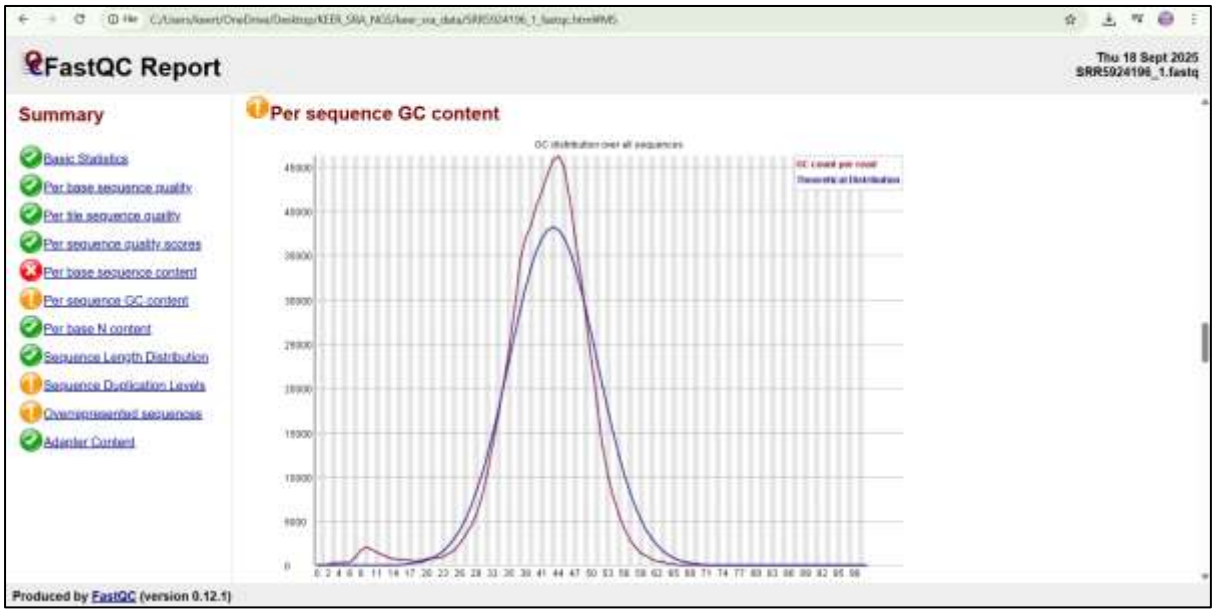


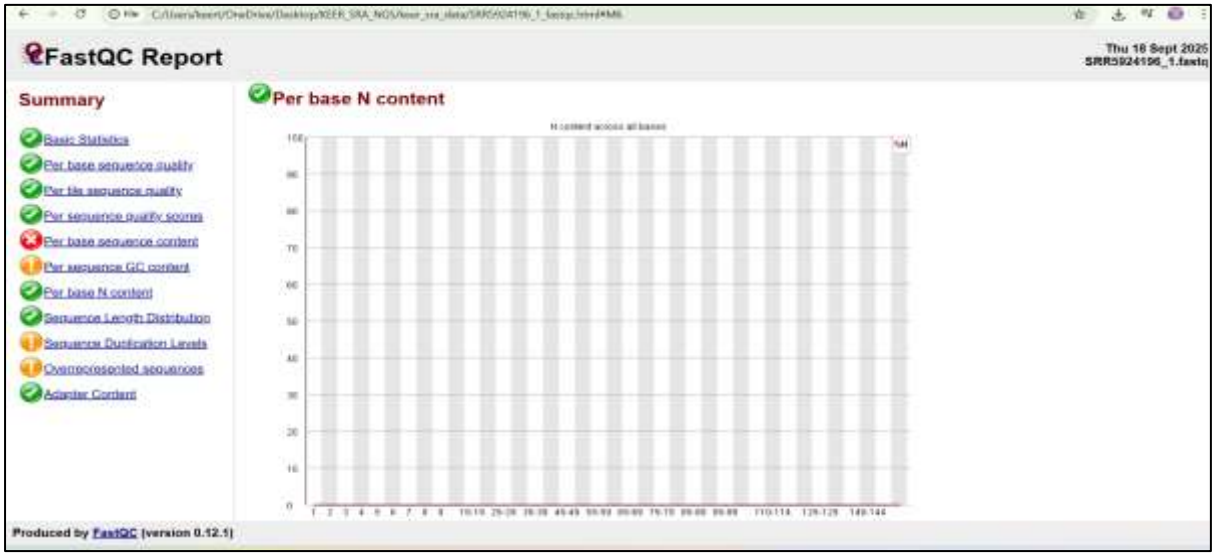

FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per base sequence quality

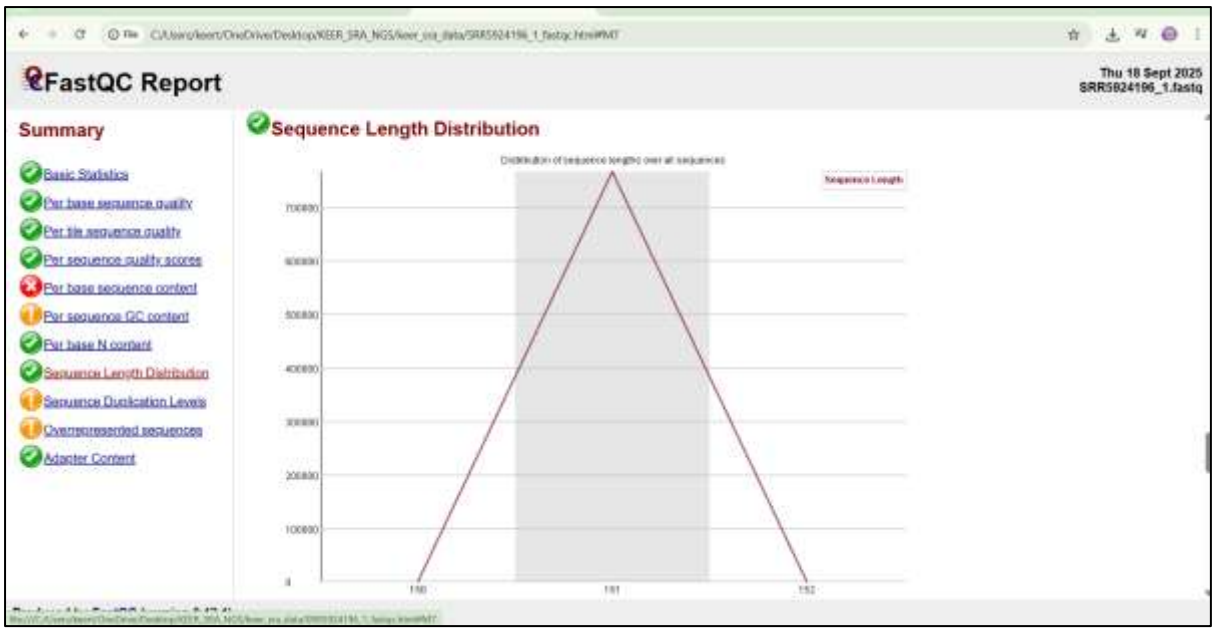

FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per tile sequence quality

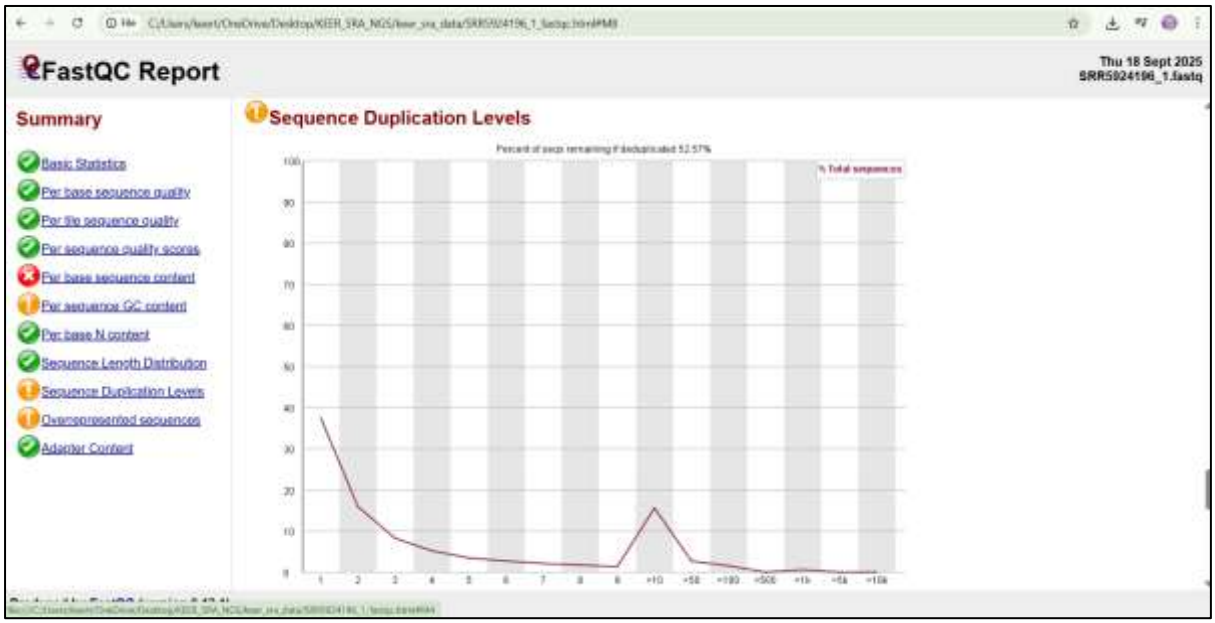

FastQC Report
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per sequence quality scores
Produced by FastQC (version 0.12.1)

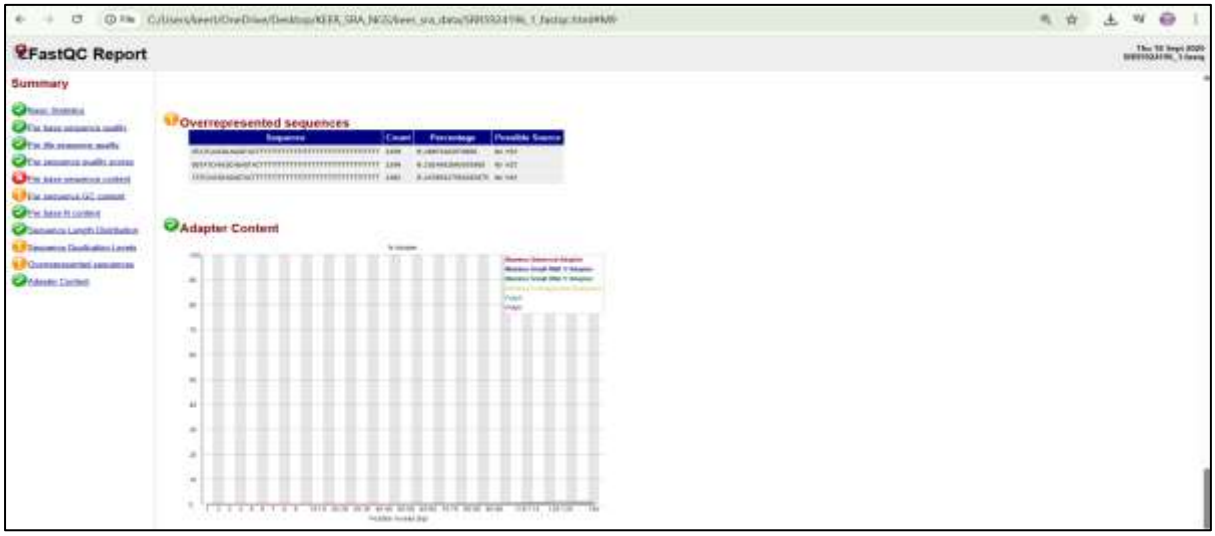

FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per base sequence content
Produced by FastQC (version 0.12.1)


FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per sequence GC content
Produced by FastQC (version 0.12.1)


FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Per base N content
Produced by FastQC (version 0.12.1)

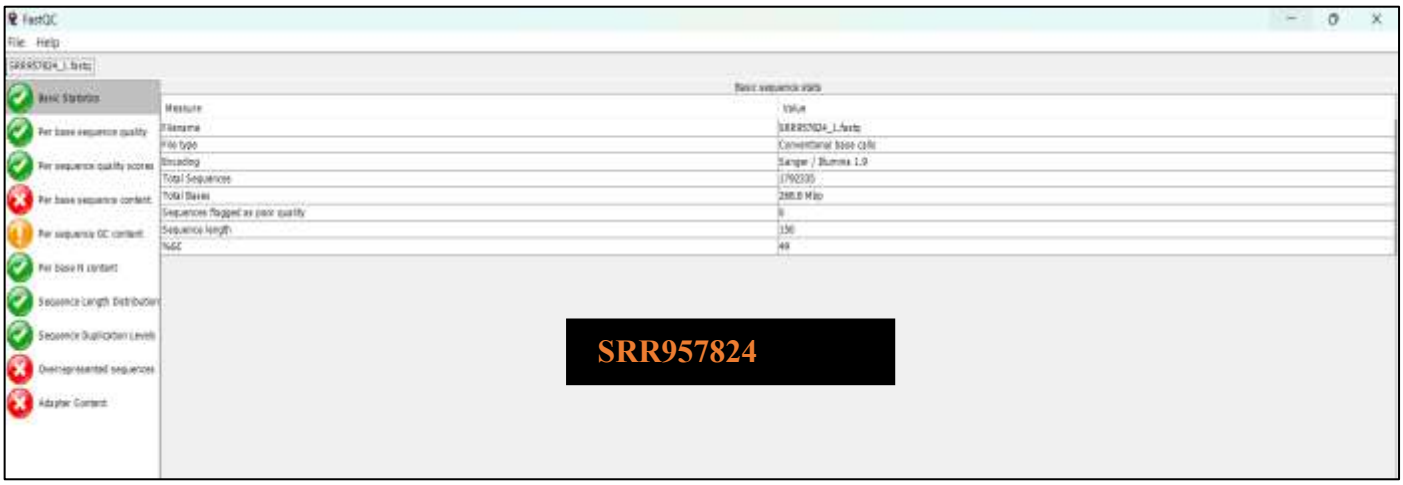

FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Sequence Length Distribution

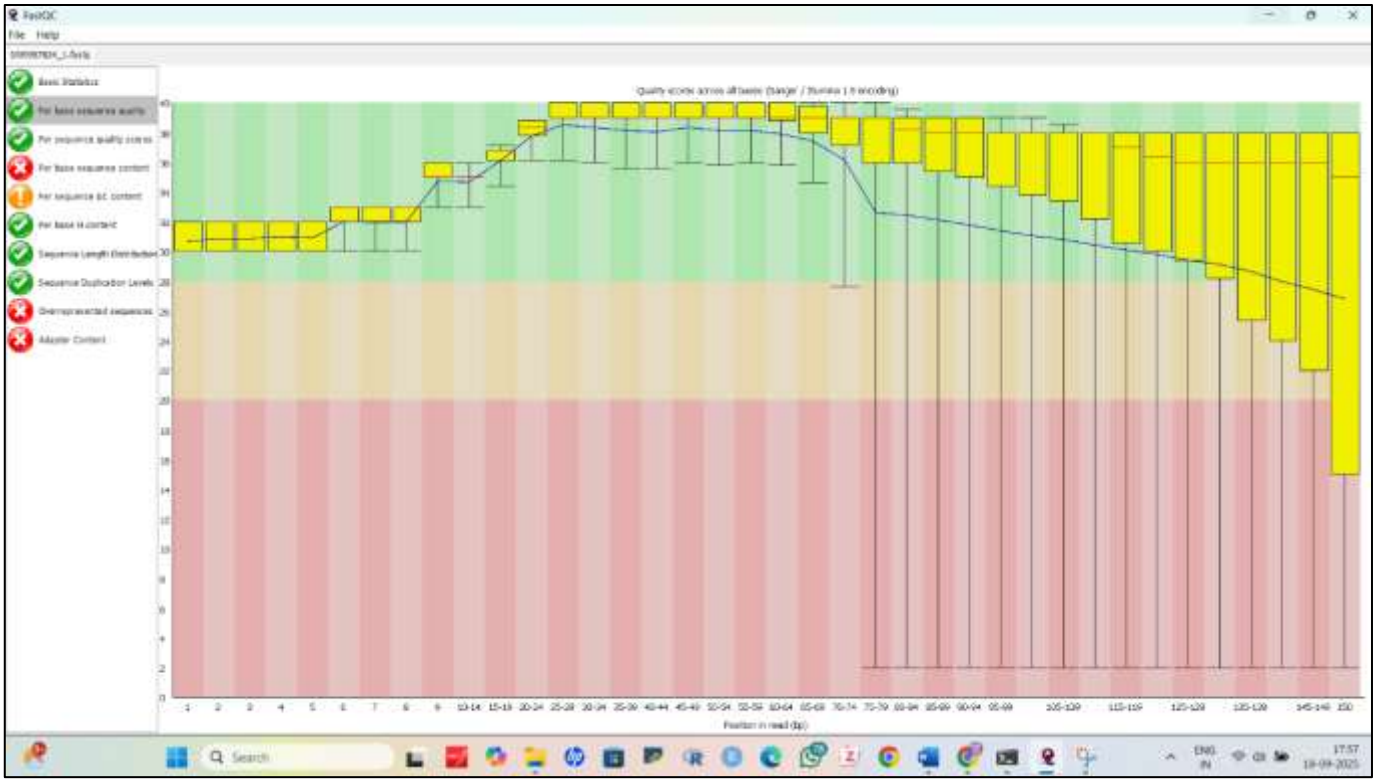

FastQC Report
Thu 18 Sept 2025
SRR5924196_1.fastq
Summary
Basic Statistics
Per base sequence quality
Per tile sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Sequence Duplication Levels

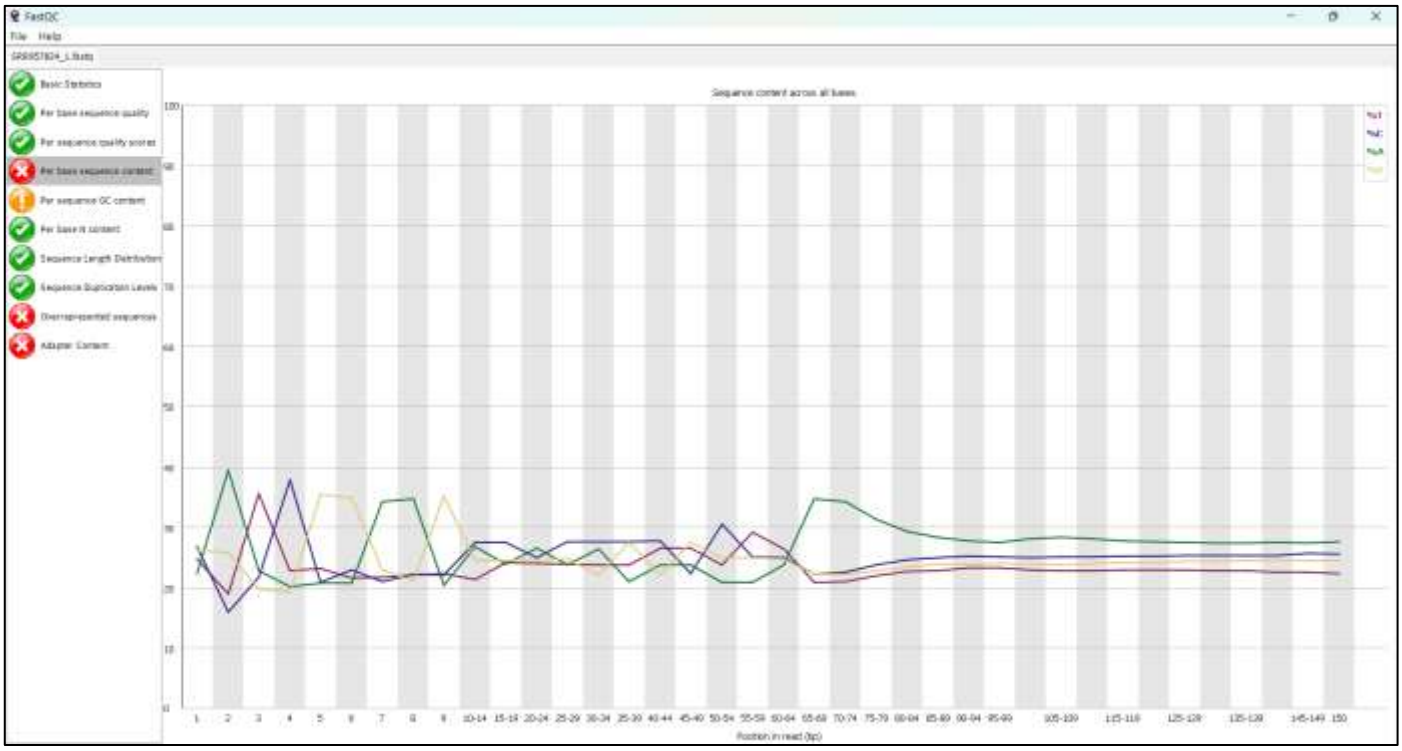

FastQC Report
Summary
Overrepresented sequences
Adapter Content

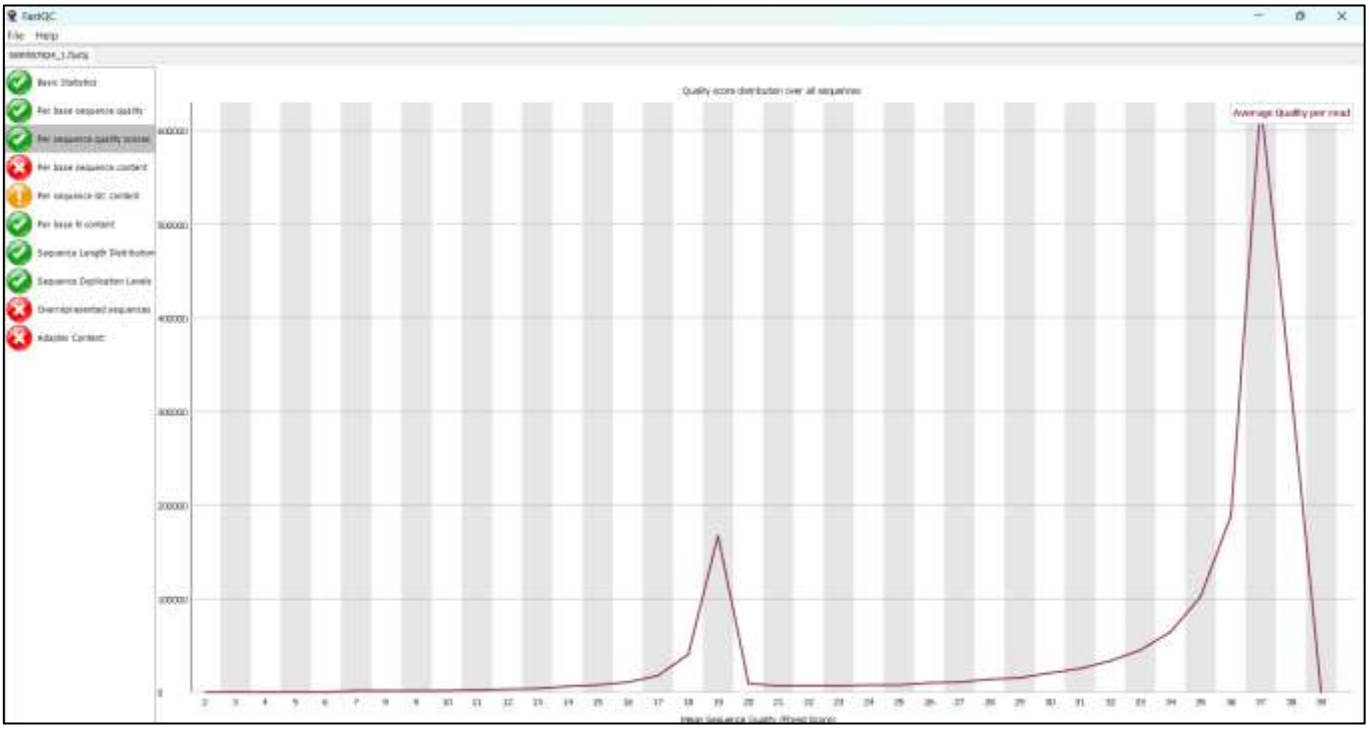
Do the same for the other one:

fasterq-dump C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data\SRR957824\SRR957824.sra -O C:\Users\keert\OneDrive\Desktop\KEER_SRA_NGS\keer_sra_data --split-files --threads 6 --progress



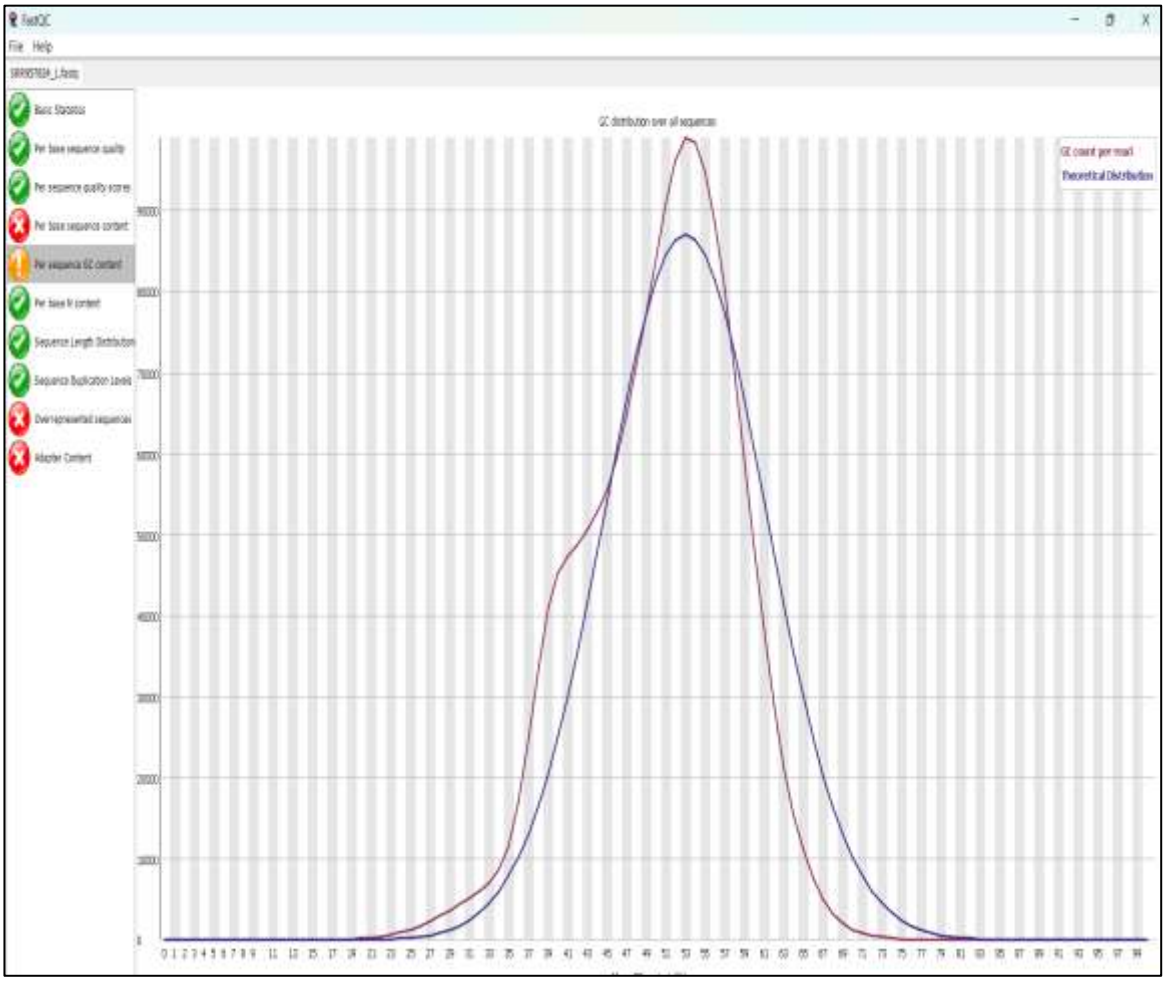


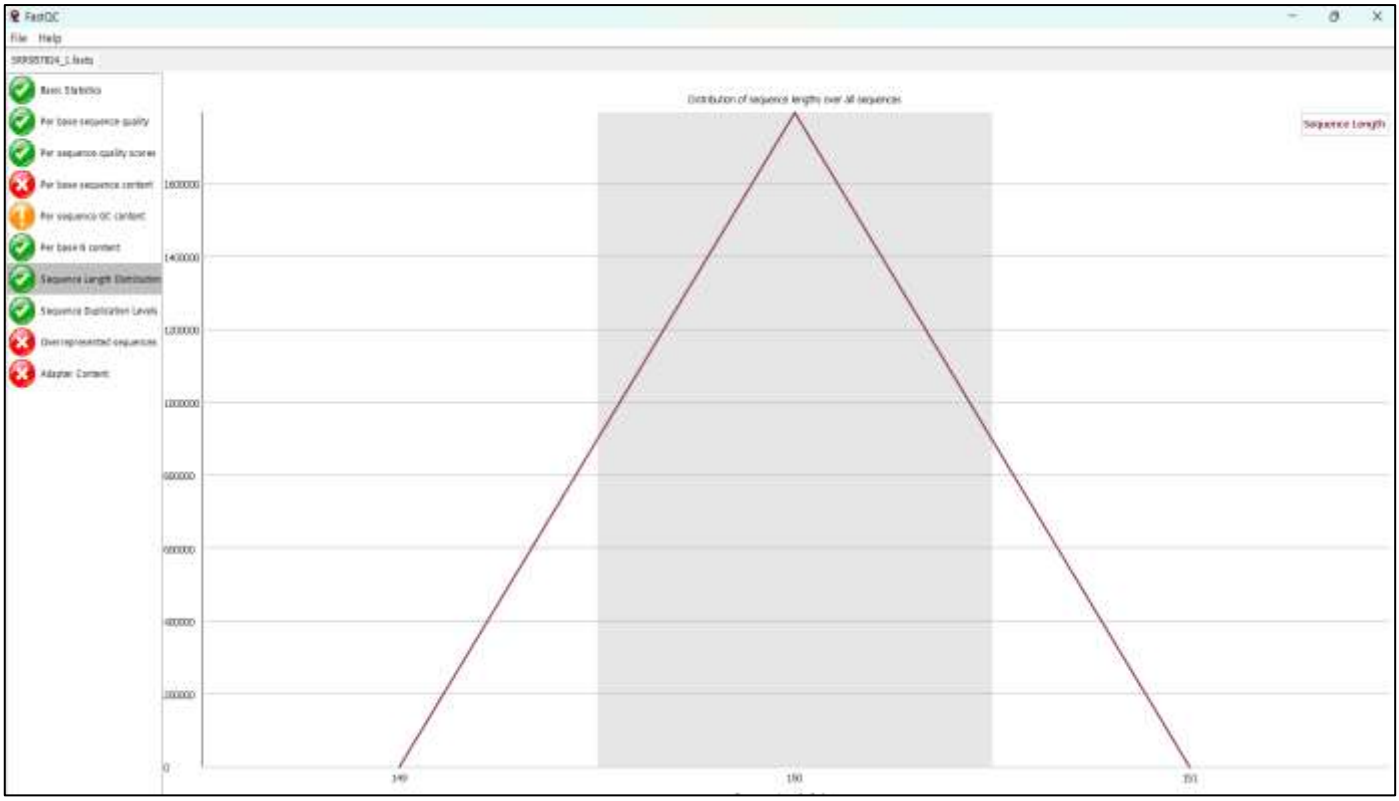

FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Sequence content across all bases
Position in read (bp)

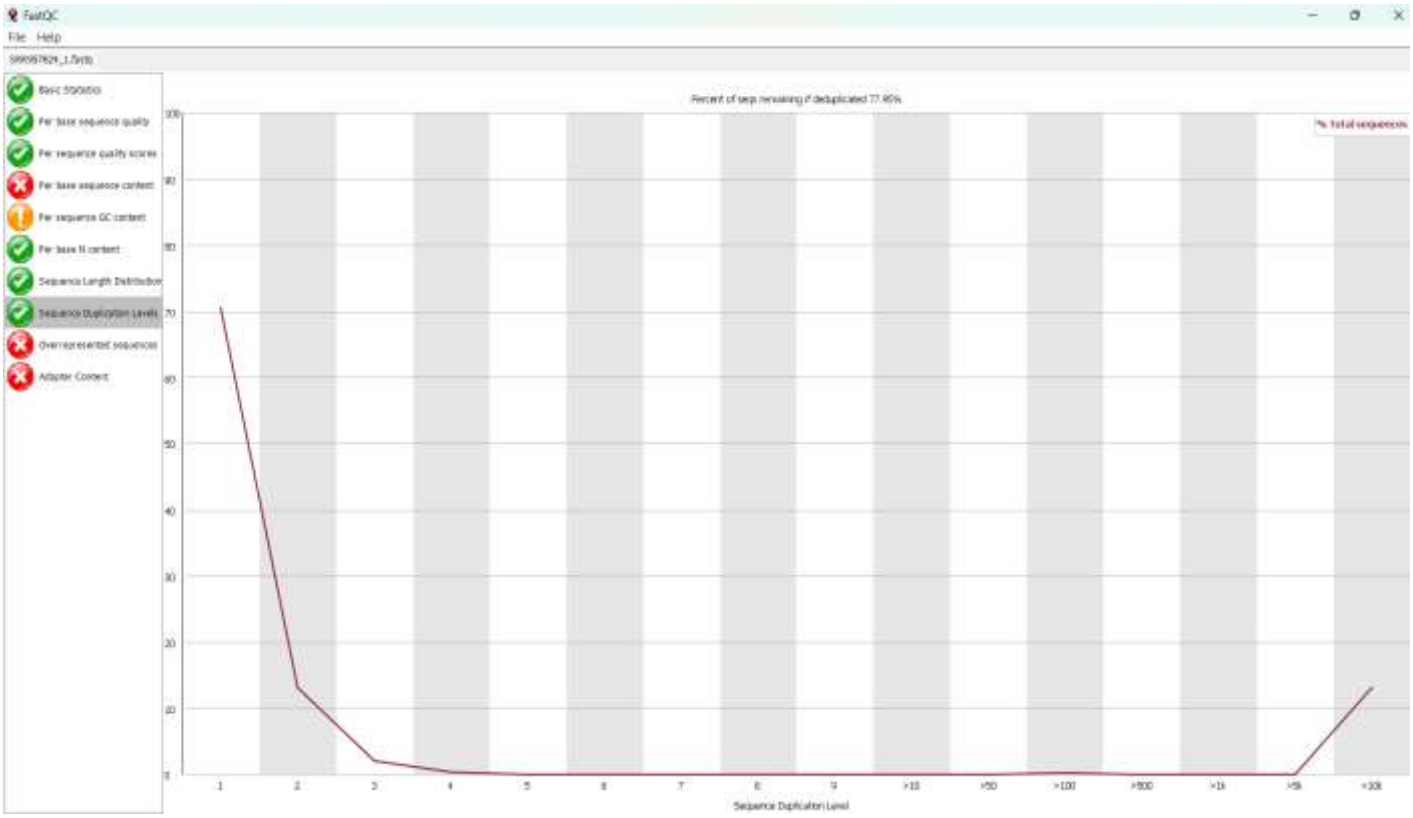

FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Quality scores distribution over all sequences
Average Quality per read

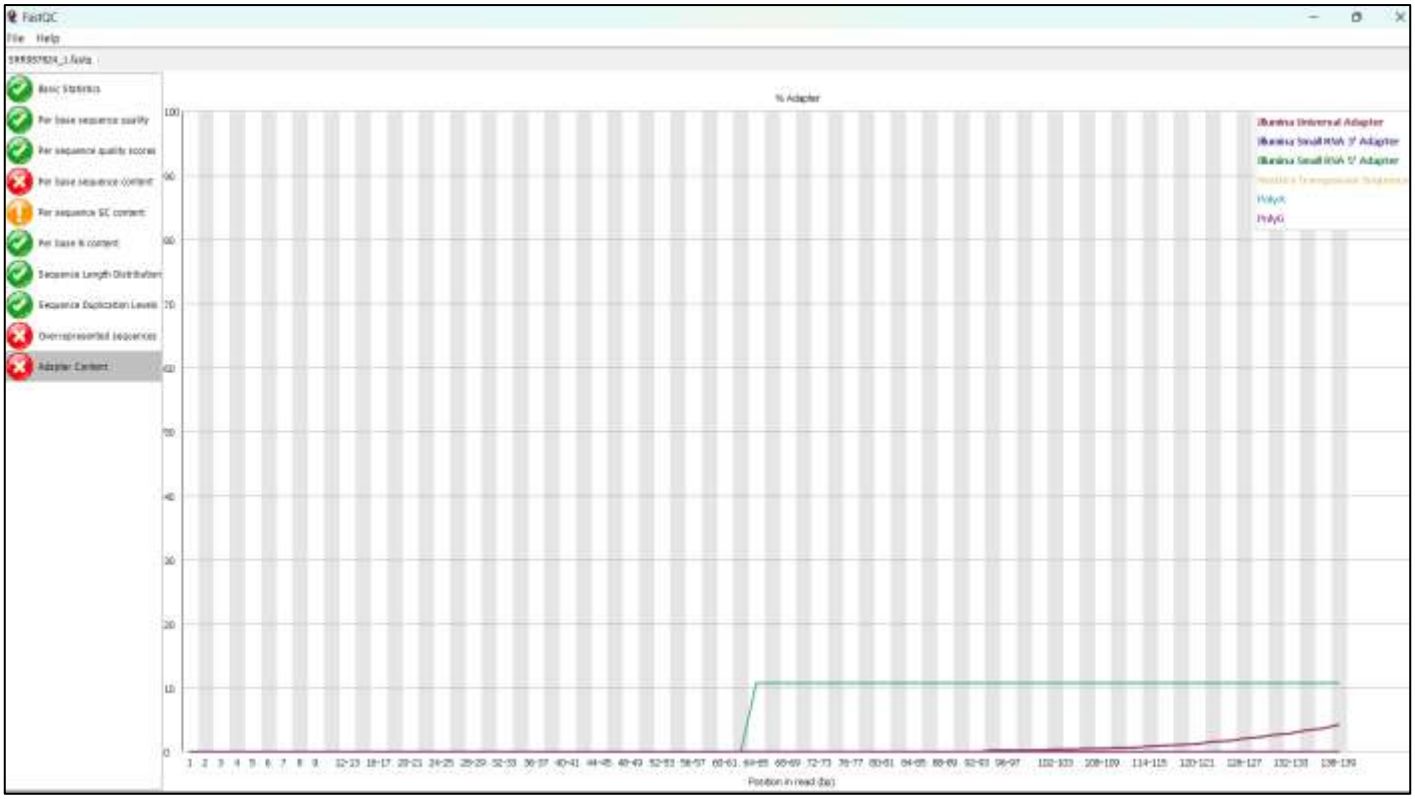
PER SEQUENCE GC CONTENT

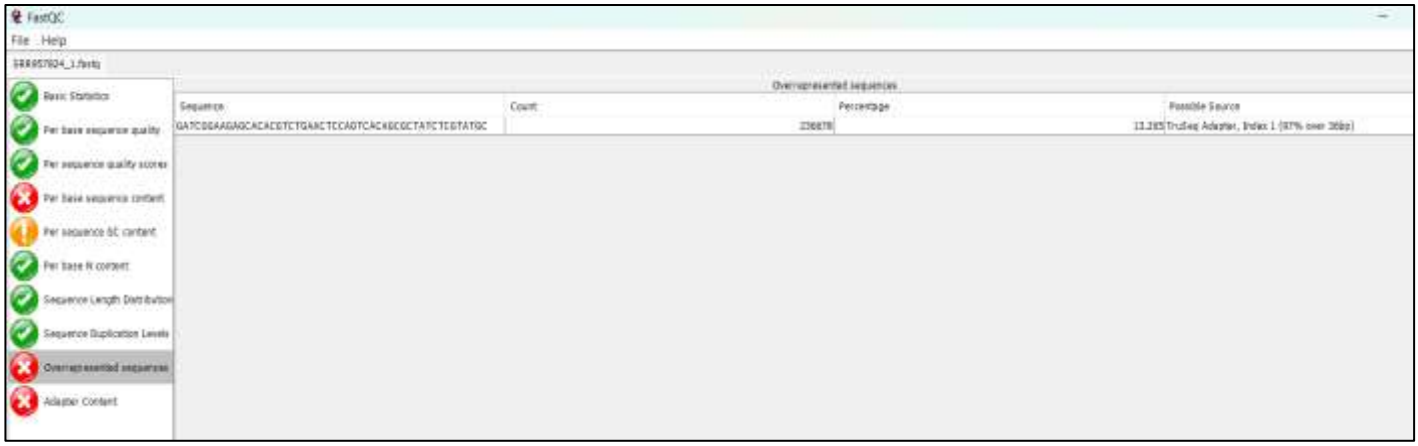
In a normal random library, you would expect to see a roughly normal distribution of GC content where the central peak corresponds to the overall GC content of the underlying genome. Since we don't know the GC content of the genome the modal GC content is calculated from the observed data and used to build a reference distribution. An unusually shaped distribution could indicate a contaminated library or some other kinds of biased subset. A normal distribution which is shifted indicates some systematic bias which is independent of base position. If there is a systematic bias which creates a shifted normal distribution then this won't be flagged as an error by the module since it doesn't know what your genome's GC content should be.

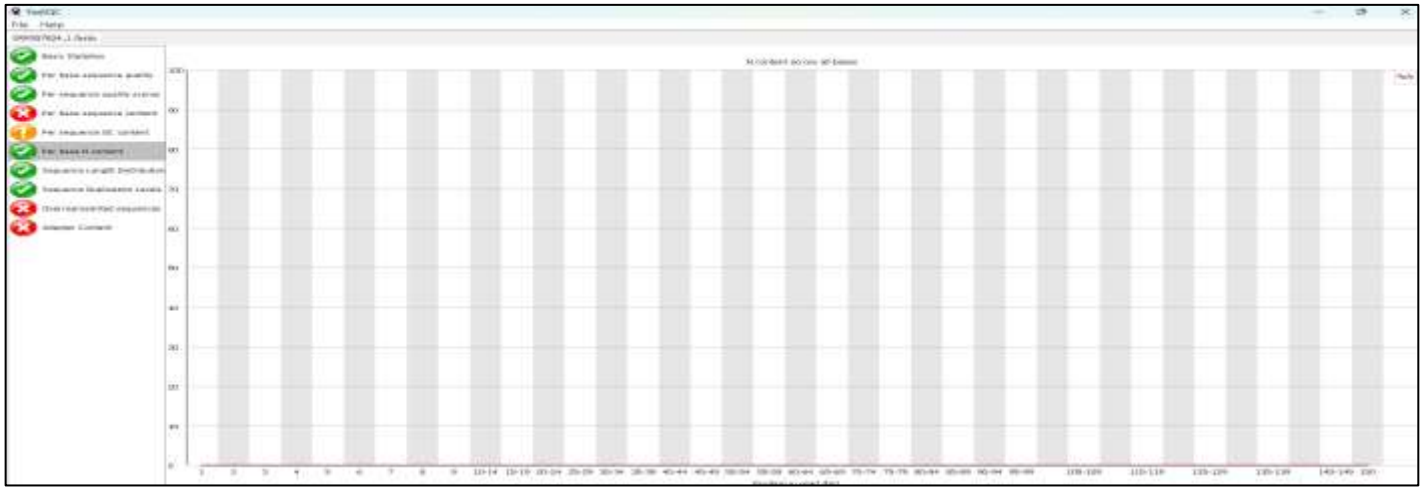



FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Distribution of sequence lengths over all sequences
Sequence Length
1600000
1400000
1200000
1000000
800000
600000
400000
200000
0
149
150
151


FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Percent of seqs remaining if deduplicated 77.95%
% total sequences
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6
7
8
9
>10
>50
>100
>500
>1k
>5k
>10k
Sequence Duplication Level


FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
% Adapter
Illumina Universal Adapter
Illumina Small RNA 3' Adapter
Illumina Small RNA 5' Adapter
PolyA
PolyG
Position in read (bp)


FastQC
File Help
Basic Statistics
Per base sequence quality
Per sequence quality scores
Per base sequence content
Per sequence GC content
Per base N content
Sequence Length Distribution
Sequence Duplication Levels
Overrepresented sequences
Adapter Content
Overrepresented sequences
Sequence
Count
Percentage
Possible Source
GATCGGAAGAGCACACGTCTGAACTCCAGTCACNGCGCTATCTCGTATGC
236878
13.285
TruSeq Adapter, Index 1 (97% over 36bp)



PRACTICE-3

OPTIMIZING SEQUENCING DATA: ADAPTER TRIMMING WITH TRIMMOMATIC

OBJECTIVE: -

Trimmomatic is a flexible, efficient, and easy-to-use tool for trimming adapters from highthroughput sequencing data. The primary objectives of using Trimmomatic for adapter trimming are:

1. **Remove Adapter Sequences**: Identify and remove sequences introduced during library preparation to prevent them from interfering with downstream analyses.
2. **Quality Control**: Enhance the quality of the data by trimming low-quality bases from the ends of reads and filtering out reads that do not meet certain quality thresholds.
3. **Improve Mapping Accuracy**: Increase the accuracy of read mapping by removing adapter contamination, which can cause incorrect alignments and affect downstream analyses such as variant calling or expression quantification.
4. **Optimize Data for Analysis**: Ensure that the reads used for analysis are of the highest quality possible, leading to more reliable and reproducible results.

SOFTWARE USED:-

Trimmomatic(http://www.usadellab.org/cms/?page=trimmomatic)

```
               3 Dir(s)  21,674,266,624 bytes free

C:\Users\keert\OneDrive\Desktop\SUBJECTS 4TH YR\NGS\Tri\Trimmomatic-0.39\Trimmomatic-0.39>java -jar trimmomatic-0.39.ja
put_SRR5924196_2_trimmed.fq.gz" ILLUMINACLIP:"adapters/TruSeq3-SE.fa":2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 M
TrimmomaticSE: Started with arguments:
 SRR5924196_2_fastqc.fq.gz output_SRR5924196_2_trimmed.fq.gz ILLUMINACLIP:adapters/TruSeq3-SE.fa:2:30:10 LEADING:3 TRAI
Automatically using 1 threads
Using Long Clipping Sequence: 'AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTGTA'
Using Long Clipping Sequence: 'AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC'
ILLUMINACLIP: Using 0 prefix pairs, 2 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences
Exception in thread "main" java.io.FileNotFoundException: SRR5924196_2_fastqc.fq.gz (The system cannot find the file sp
        at java.base/java.io.FileInputStream.open0(Native Method)
        at java.base/java.io.FileInputStream.open(FileInputStream.java:213)
        at java.base/java.io.FileInputStream.<init>(FileInputStream.java:152)
        at org.usadellab.trimmomatic.fastq.FastqParser.parse(FastqParser.java:135)
        at org.usadellab.trimmomatic.TrimmomaticSE.process(TrimmomaticSE.java:197)
        at org.usadellab.trimmomatic.TrimmomaticSE.run(TrimmomaticSE.java:321)
        at org.usadellab.trimmomatic.Trimmomatic.main(Trimmomatic.java:85)

C:\Users\keert\OneDrive\Desktop\SUBJECTS 4TH YR\NGS\Tri\Trimmomatic-0.39\Trimmomatic-0.39>java -jar trimmomatic-0.39.ja
.fastq" "output_forward_paired.fq.gz" "output_forward_unpaired.fq.gz" "output_reverse_paired.fq.gz" "output_reverse_unp
ruSeq3-PE.fa":2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36
TrimmomaticPE: Started with arguments:
 SRR957824_1.fastq SRR957824_2.fastq output_forward_paired.fq.gz output_forward_unpaired.fq.gz output_reverse_paired.fq
UMINACLIP:adapters/TruSeq3-PE.fa:2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:36
Using PrefixPair: 'TACACTCTTTCCCTACACGACGCTCTTCCGATCT' and 'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'
ILLUMINACLIP: Using 1 prefix pairs, 0 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences
Quality encoding detected as phred33
Input Read Pairs: 1792335 Both Surviving: 1432723 (79.94%) Forward Only Surviving: 348117 (19.42%) Reverse Only Survivi
)
TrimmomaticPE: Completed successfully
```

Analysing the report of FastQC after adapter trimming using trimmomatic :-

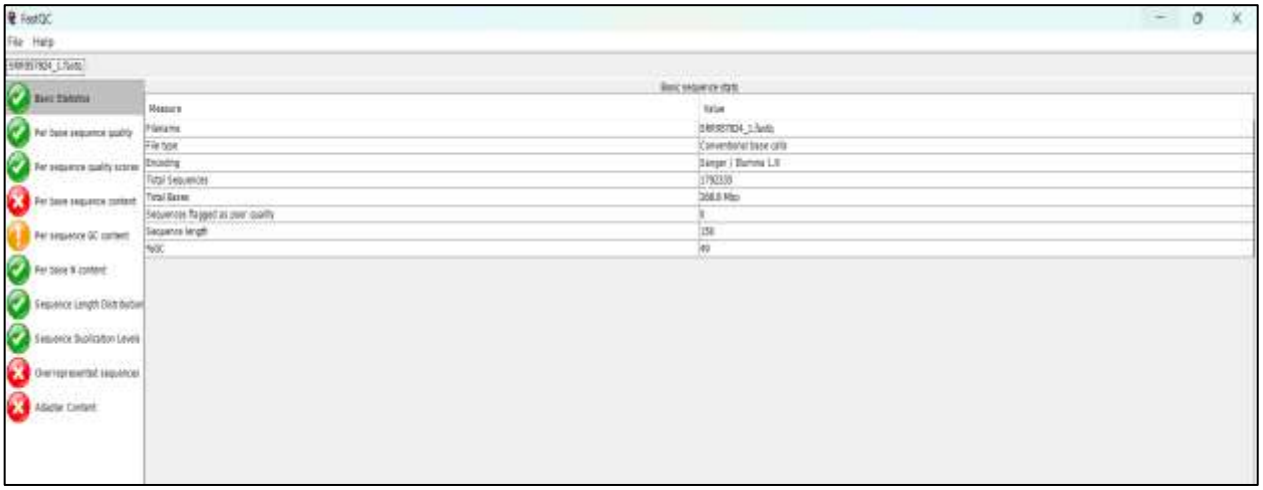
The different analyses (modules) performed for the FastQC report are summarized on the left of the screen, for each FASTQ file used for the analysis.

Basic Statistics

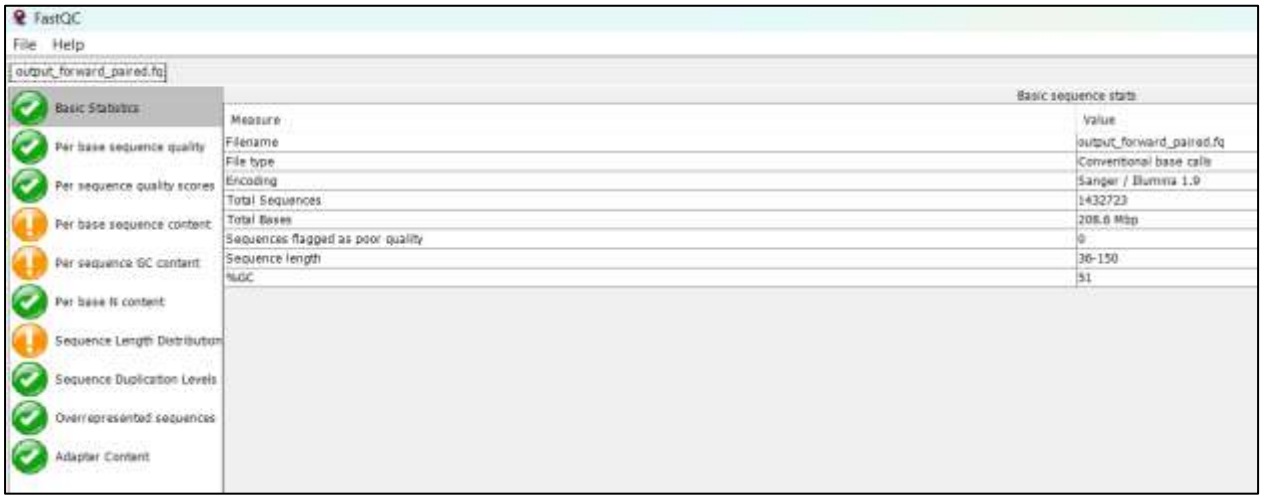
Simple information about input FASTQ file: its name, type of quality score encoding, total number of reads, read length and GC content.

- The Basic Statistics section of a FastQC report provides a quick overview of the sequencing data quality. Here are the key metrics included:
- Total Sequences: This indicates the total number of sequences in the dataset, helping to assess the size of the sample.
- Filtered Sequences: If applicable, this shows how many sequences were flagged for filtering, which is crucial for understanding data quality.
- Sequence Length: FastQC reports the shortest and longest sequence lengths, providing insight into the uniformity of the data.

BEFORE TRIMMING



AFTER TRIMMING



The Per Base Sequence Quality analysis is essential for ensuring that the sequencing data is reliable. By examining the quality scores, researchers can make informed decisions about data processing, such as whether to trim low-quality bases or to contact the sequencing facility if significant issues are detected. This module serves as a foundational step in quality control, guiding subsequent analyses and interpretations. For each position a Box Whisker type plot is drawn. The elements of the plot are as follows:

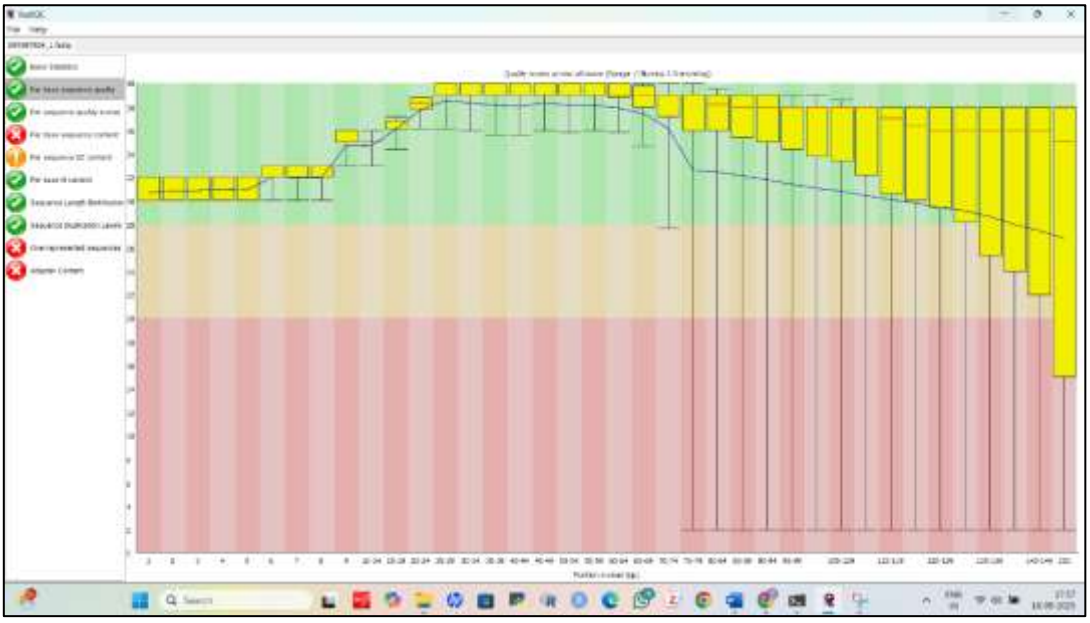
•The central red line is the median value

•The yellow box represents the inter-quartile range (25-75%)

•The upper and lower whiskers represent the 10% and 90% points

•The blue line represents the mean quality.

BEFORE TRIMMING

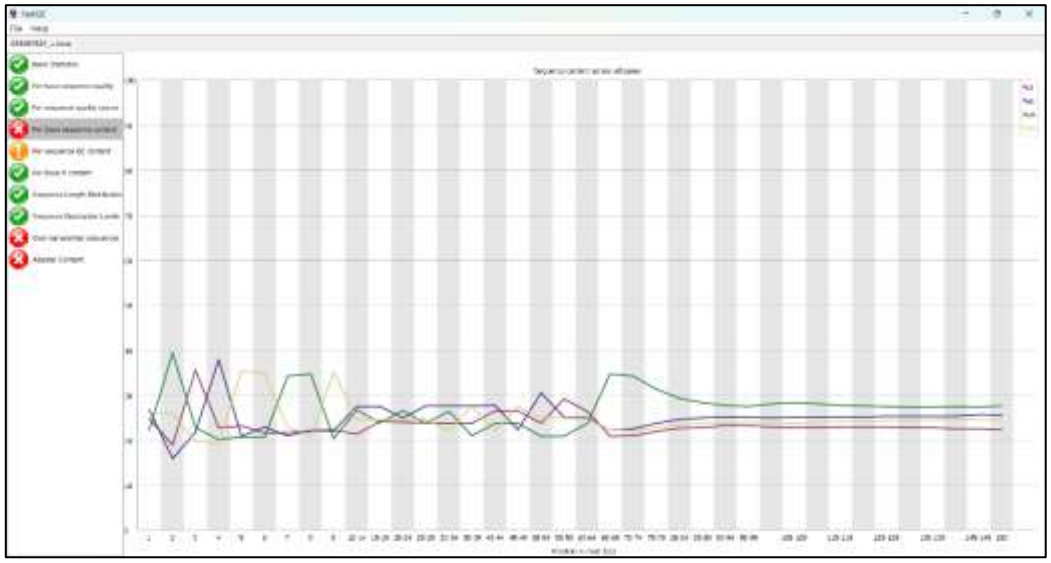


AFTER TRIMMING

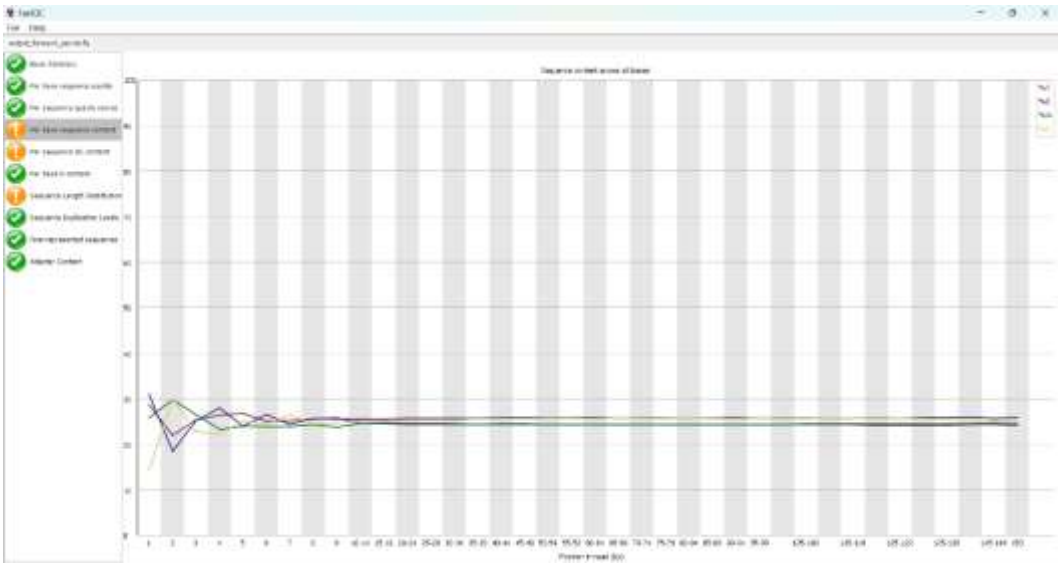
Per Base Sequence Content:

- The Per Base Sequence Content module in FastQC evaluates the proportion of each nucleotide (A, T, C, G) at every position across all reads.
- Ideally, the nucleotide distribution should be relatively flat, reflecting the expected genome composition. Significant imbalances can indicate issues such as biased sequence composition or contamination from overrepresented sequences like adapter dimers.
- Systematic deviations, especially at the 3' end, may suggest the presence of adapter sequences, affecting downstream analyses.
- A balanced sequence content is crucial for reliable mapping and variant calling, making it essential to address any identified issues before further analysis

BEFORE TRIMMING



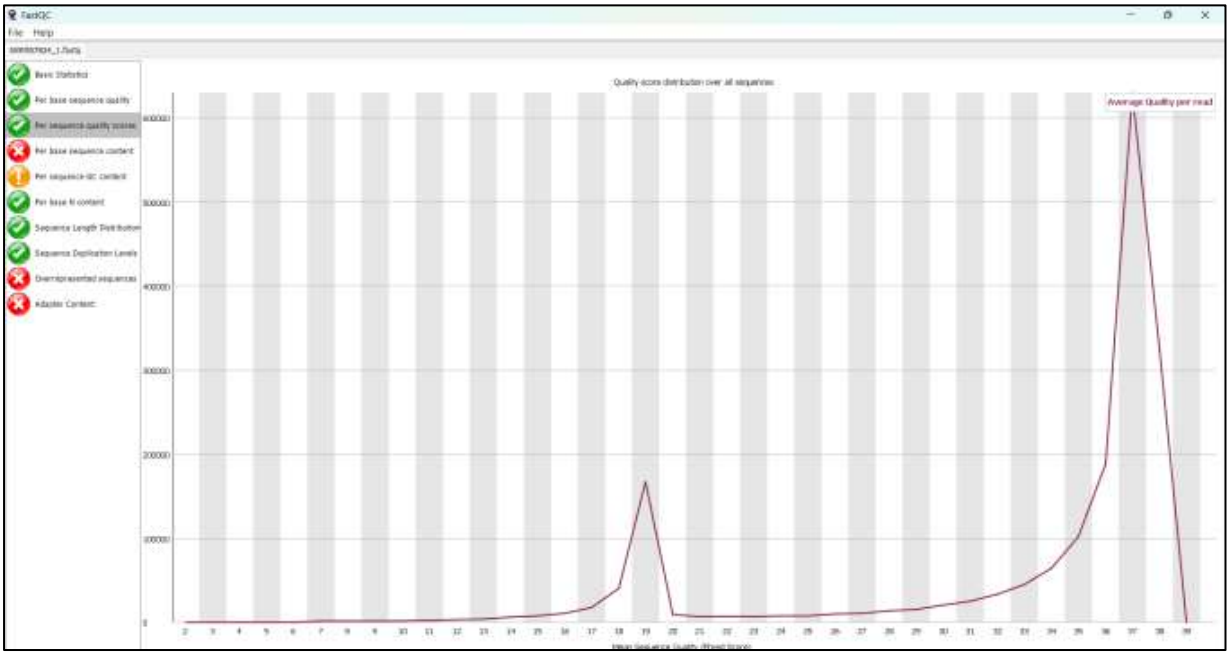
AFTER TRIMMING



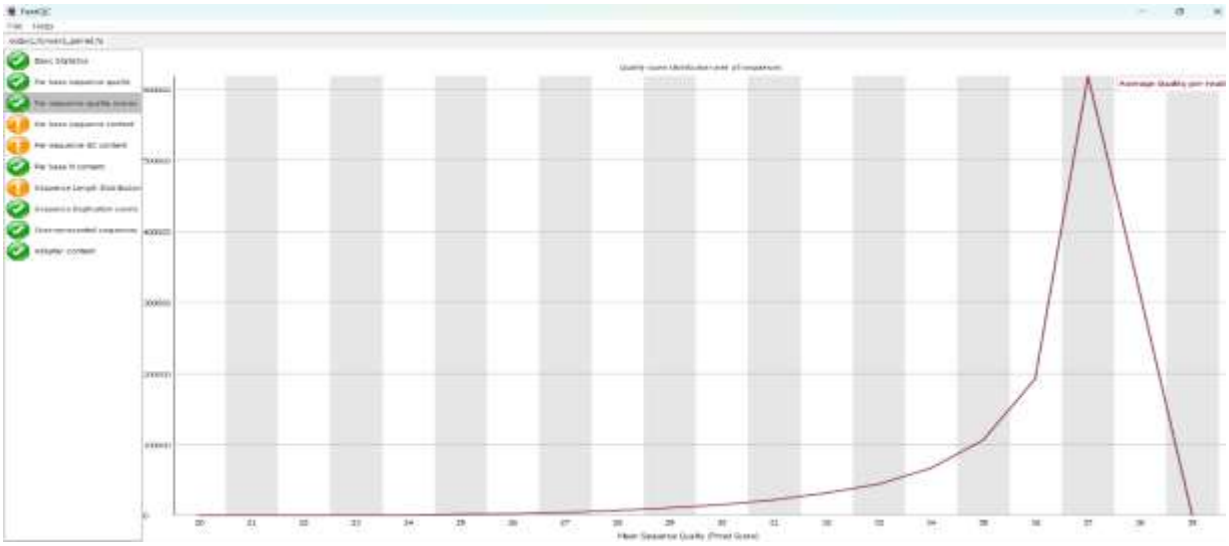
Per Sequence Quality Scores:

- The Per Sequence Quality Scores in FastQC provide an overview of sequencing read quality, helping to identify potential issues.
- This module displays the distribution of quality scores across the dataset, highlighting any sequences with low quality that may indicate systematic problems, particularly in long runs.
- The results are visualized in a box plot, with color coding to differentiate between good, reasonable, and inferior quality scores.
- A noticeable drop in quality at the end of reads suggests the need for trimming low-quality bases to improve overall data quality.
- Evaluating these scores alongside other metrics is essential for a comprehensive assessment of sequencing quality and for guiding data preprocessing decisions.

BEFORE TRIMMING



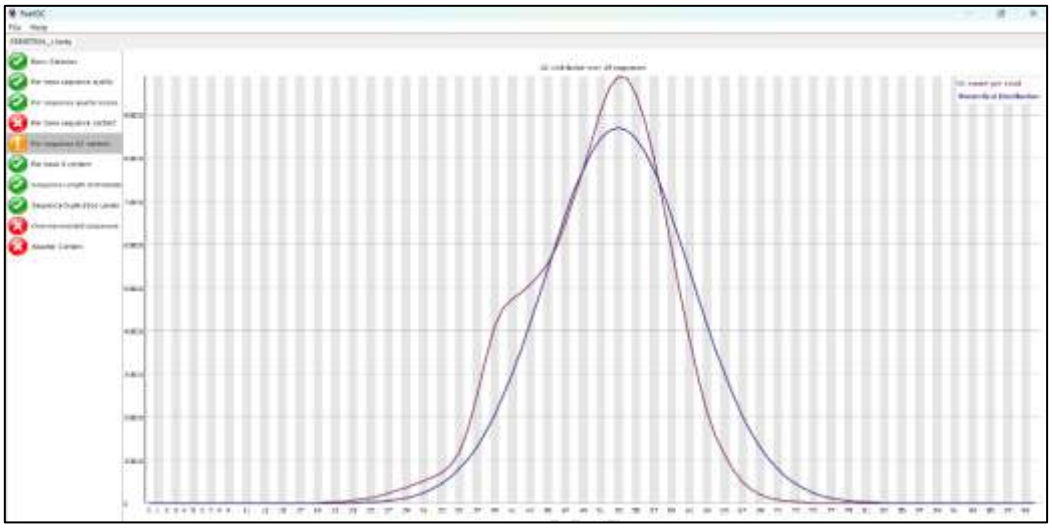
AFTER TRIMMING



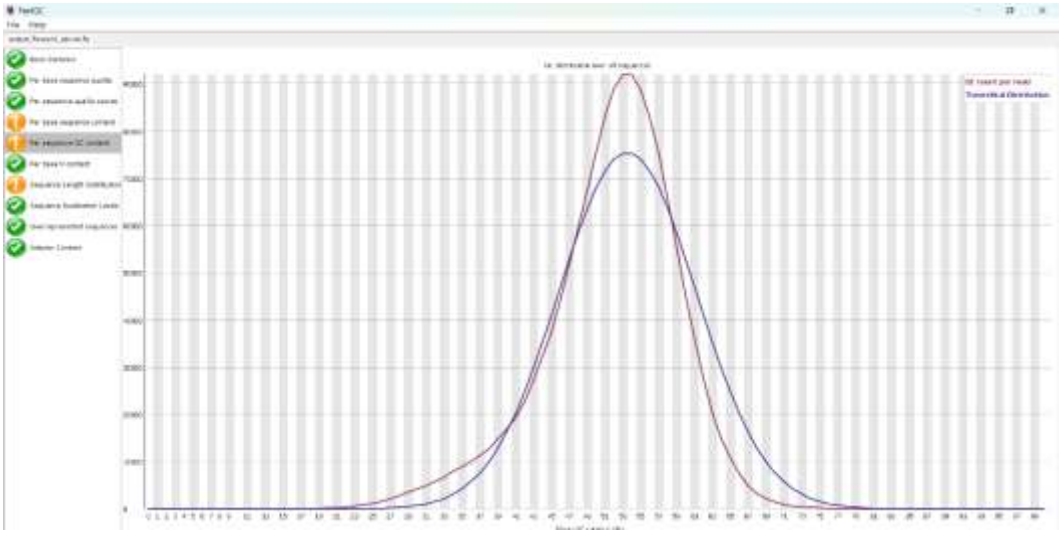
Per Base GC Content:

- The Per Base GC Content plot in FastQC shows the percentage of G and C bases at each position in the sequencing reads.
- This plot helps identify biases in the sequencing data, such as GC-rich or GC-poor regions.
- Ideally, the GC content should be evenly distributed across the reads, with no significant deviations from the average GC content of the genome. Significant deviations can indicate issues with library preparation, sequencing bias, or contamination**.**
- In FastQC, the Per Base Sequence Content module raises a warning if the sum of deviations from the normal distribution exceeds 15% of the reads. A more severe failure occurs if this sum surpasses 30% of the reads, suggesting significant deviations that could compromise the reliability of the sequencing data. Monitoring these thresholds is essential for ensuring high-quality results.

BEFORE TRIMMING



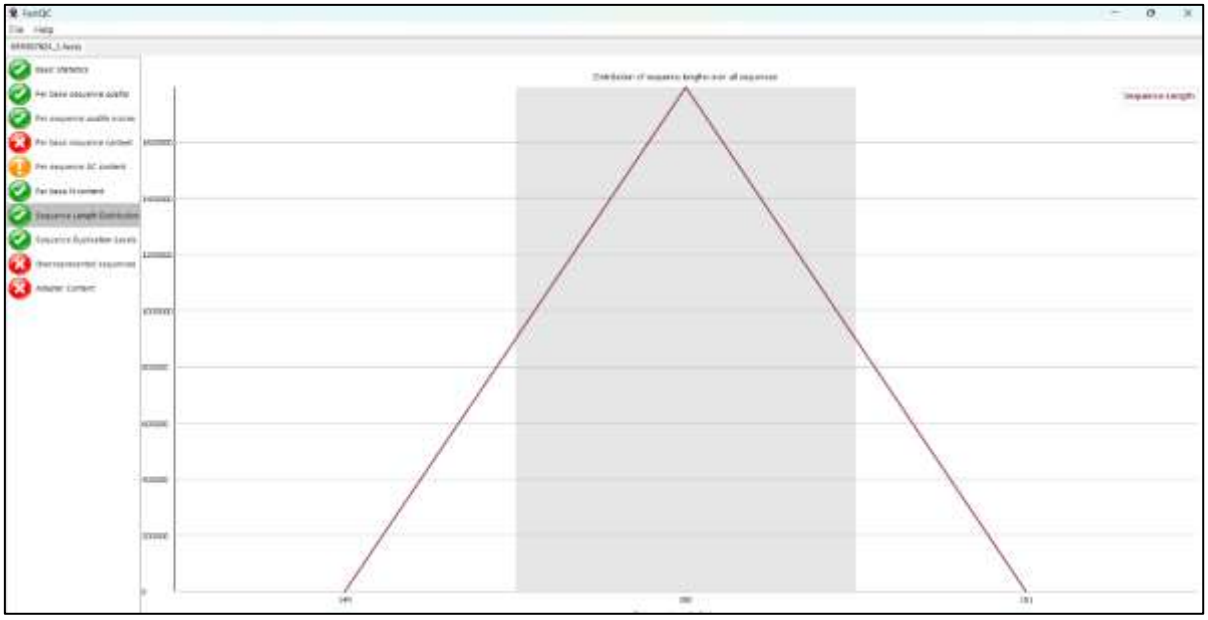
AFTER TRIMMING



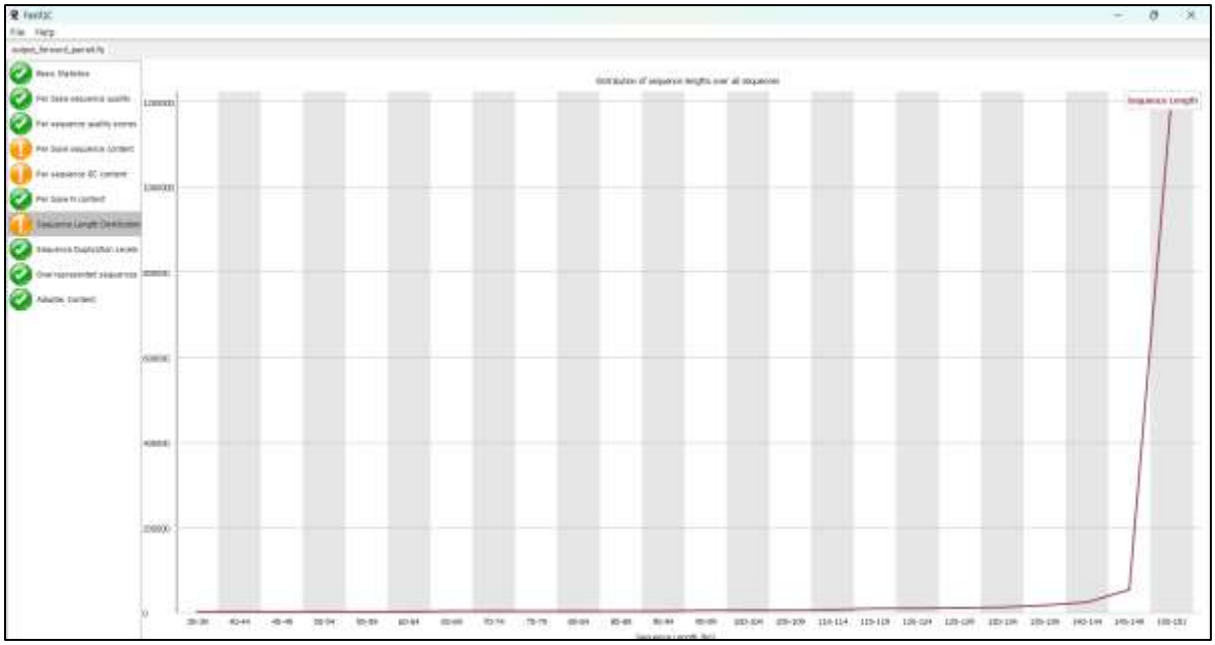
Sequence Length Distribution:

- The Sequence Length Distribution module in FastQC provides a graphical representation of the lengths of sequences in a dataset. This analysis helps assess the quality of the sequencing run by displaying the frequency of different sequence lengths.
- Ideally, the distribution should show a clear peak at the expected read length, indicating consistent and high-quality sequencing. Significant deviations, such as a broad range of lengths or unexpected shorter sequences, may suggest issues with the library preparation or sequencing process.
- Monitoring sequence length distribution is crucial for ensuring the integrity and usability of the sequencing data for downstream applications.

BEFORE TRIMMING



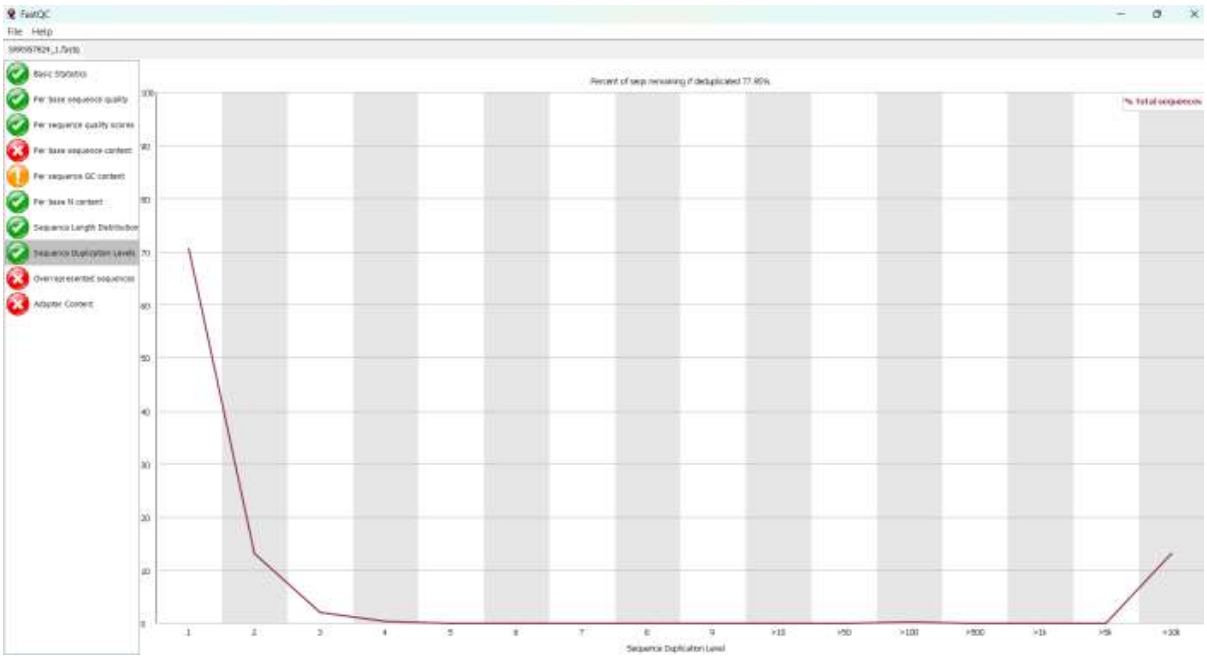
AFTER TRIMMING



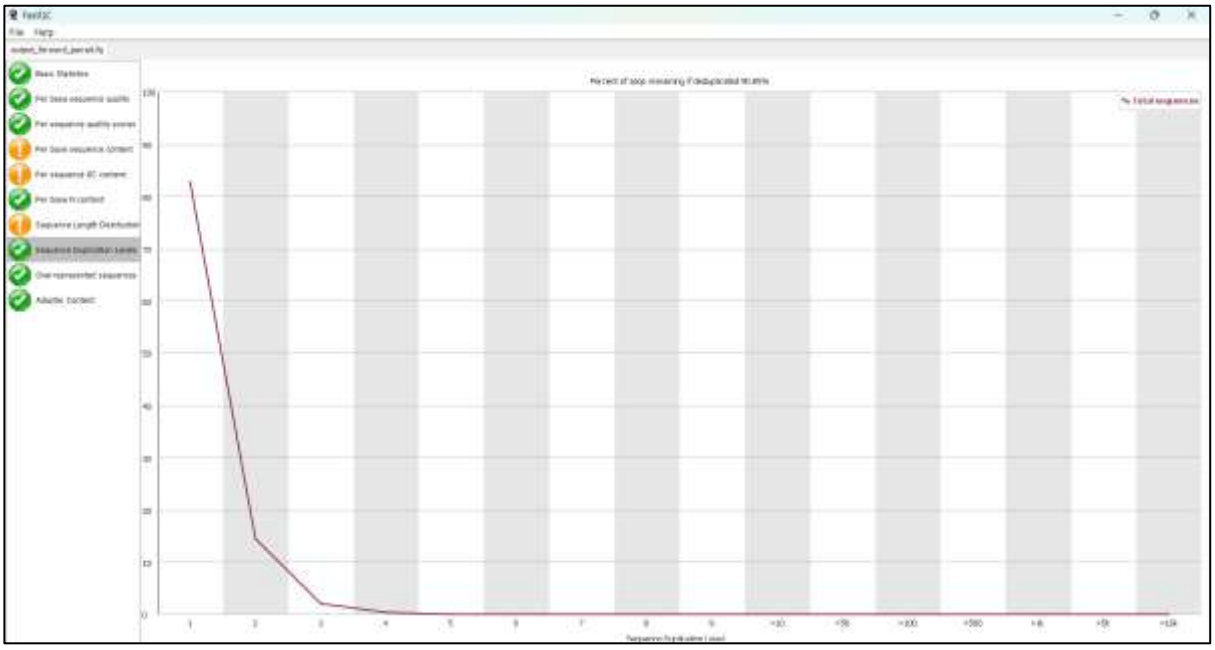
Duplicate Sequences:

- The Duplicate Sequences module in FastQC identifies and quantifies the presence of duplicate reads in a sequencing dataset.
- Duplicate sequences arise during library preparation or PCR amplification and can lead to biased estimates of variant frequency and gene expression levels.
- This module provides insights into the extent of duplication, with a focus on the percentage of total reads that are identified as duplicates.
- A high level of duplicates may indicate potential issues with the library preparation process or insufficient sequencing depth. Monitoring duplicate sequences is essential for ensuring accurate data analysis and interpretation in genomic studies**.**

BEFORE TRIMMING



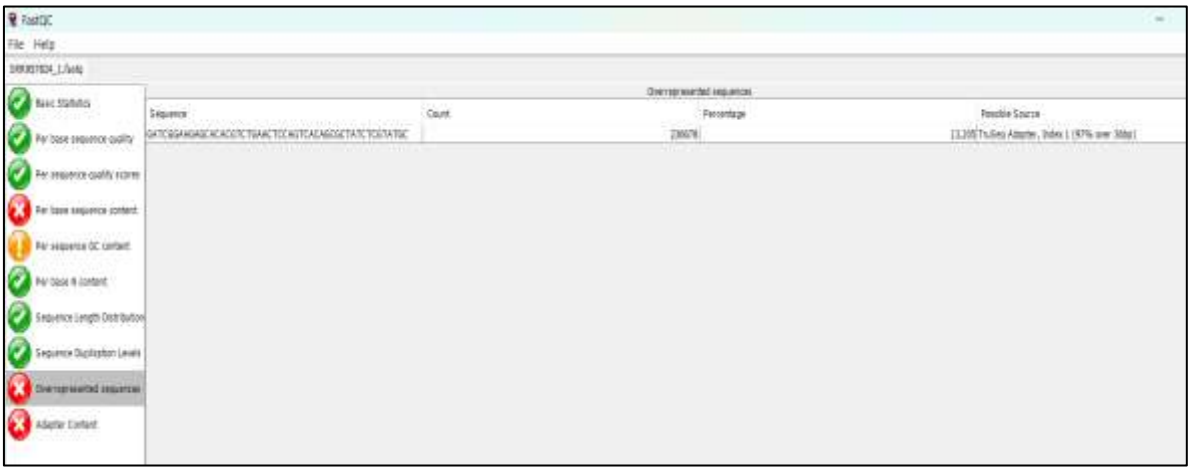
AFTER TRIMMING



Overrepresented Sequences:

- Overrepresented sequences in FastQC indicate that certain sequences appear more frequently than expected in a sequencing dataset, which can signal biological significance or potential contamination.
- FastQC analyzes the first 200,000 reads to identify sequences constituting over 0.1% of the total, comparing them against a database of common contaminants.
- These sequences can stem from various sources, such as adapter sequences, primer dimers, or specific biological entities like rRNA.
- Recognizing and addressing overrepresented sequences is crucial, as they can skew results in downstream analyses, making their identification and potential removal essential for ensuring data quality and reliability.

BEFORE TRIMMING

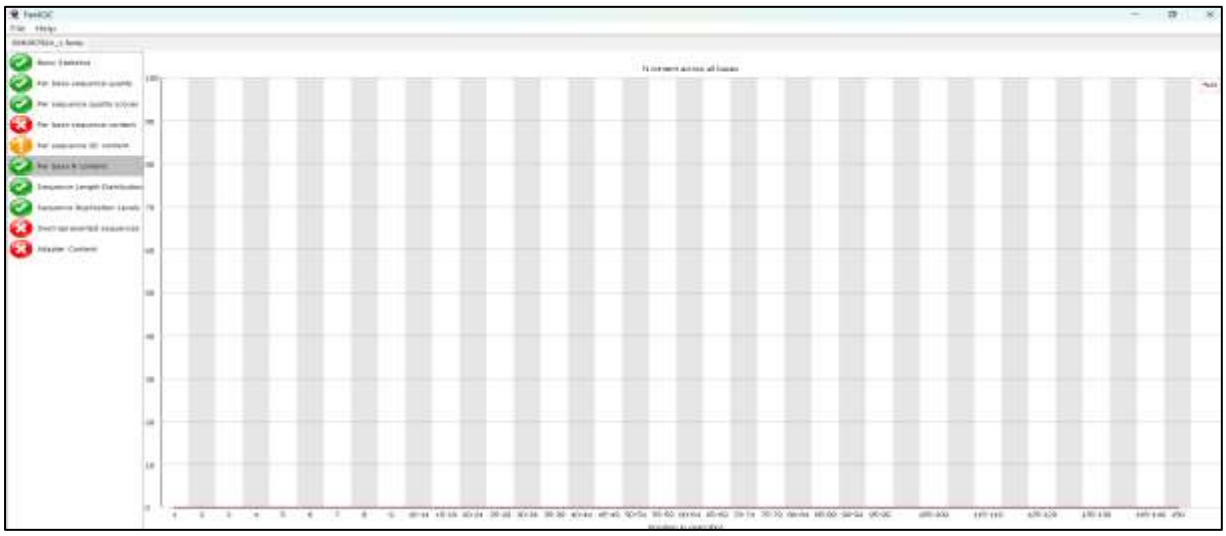


AFTER TRIMMING

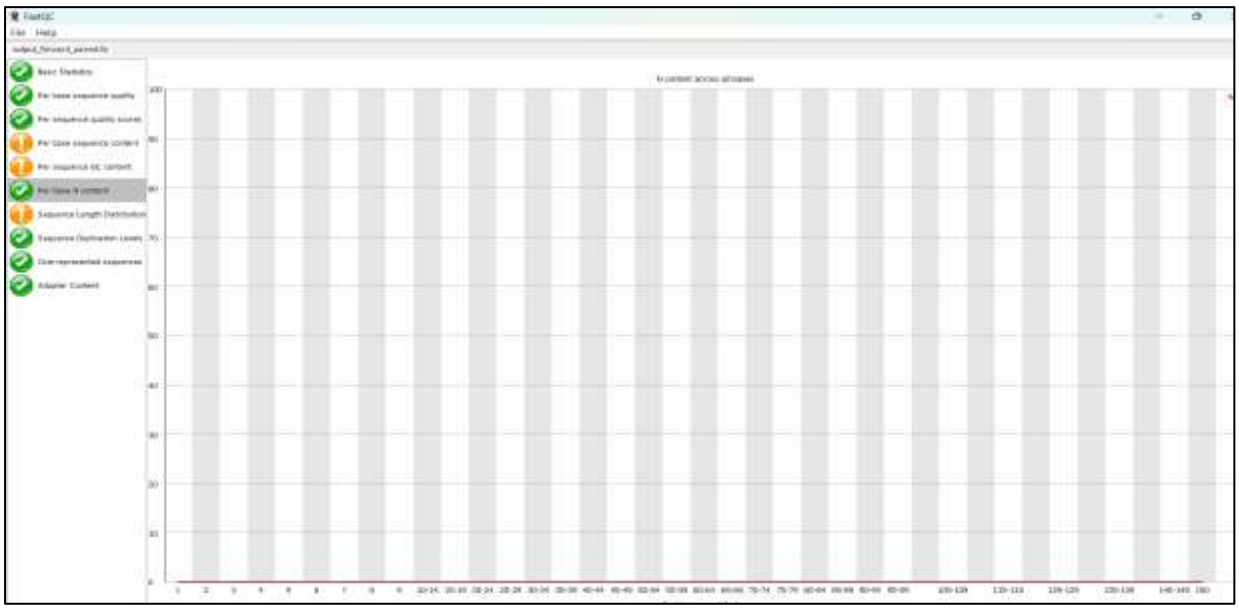
Per Base N Content:

- The Per Base N Content module in FastQC analyses the proportion of ambiguous bases (N) at each position across all reads in a sequencing dataset.
- An increase in N content can indicate issues such as poor sequencing quality or problems during the library preparation process. Ideally, the N content should be low and consistent across all positions.
- High levels of ambiguous bases, especially towards the ends of reads, may signal inadequate sequencing quality, which can adversely affect downstream analyses.
- Monitoring N content is essential for ensuring the reliability and accuracy of sequencing data

BEFORE TRIMMING

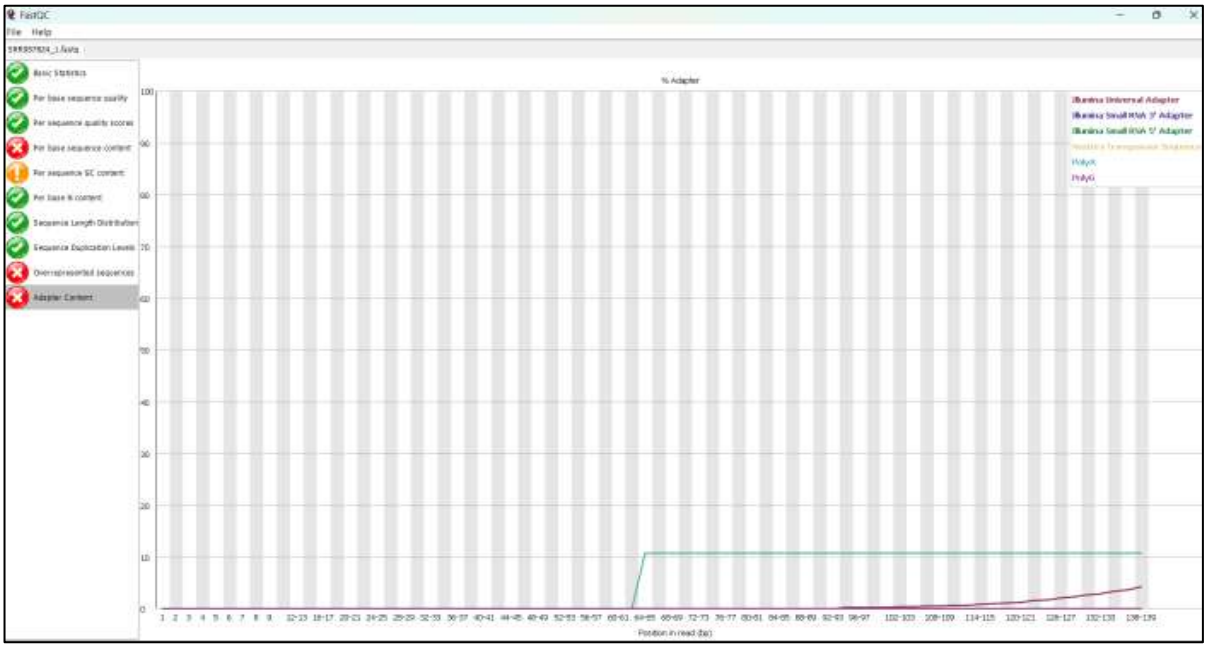


AFTER TRIMMING

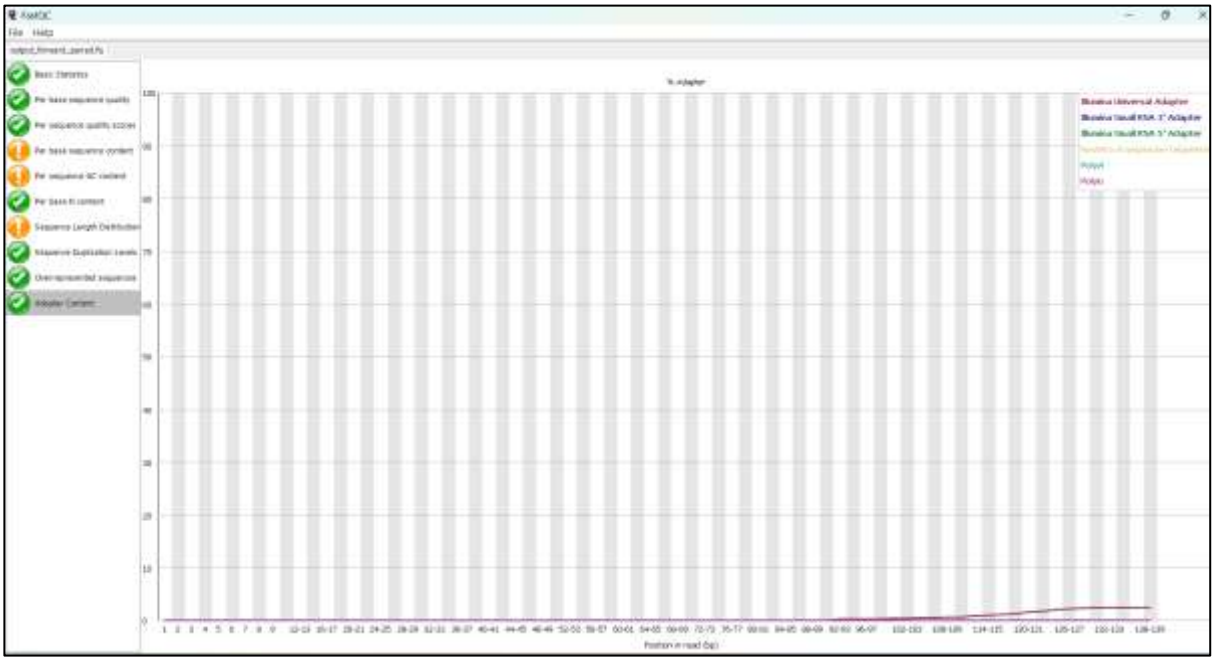


ADAPTER CONTANT

BEFORE TRIMMING



AFTER TRIMMING



PRACTICE-4:

DEG ANALYSIS USING DOWN REGULATION & DOWN REGULATION

Objective:

The objective of Differentially Expressed Gene (DEG) analysis is to identify genes whose expression levels significantly change between different experimental conditions, such as control vs treatment. By analyzing both upregulated and downregulated genes, we aim to understand the biological processes and pathways affected, revealing key molecular mechanisms, biomarkers, or potential therapeutic targets associated with the condition under study.

SOFTWARE USED:

1. DESeq2

Type: R Bioconductor package

Use: Differential expression analysis for RNA-Seq data.

Function: Statistically identifies upregulated and downregulated genes based on normalized read counts.

Website: https://bioconductor.org/packages/release/bioc/html/DESeq2.html

2. SAR Tools

Type: R pipeline

Use: Simple pipeline wrapping DESeq2 or edgeR for DEG analysis.

Function: Provides automated DEG analysis report including up/downregulated genes.

Website: https://github.com/PF2-pasteur-fr/SARTools

3. Cuffdiff (part of Cufflinks)

Type: Command-line tool

Use: RNA-Seq differential expression analysis.

Function: Identifies DEGs (both upregulated and downregulated genes) from aligned RNA-Seq data.

Website: http://cole-trapnell-lab.github.io/cufflinks/

PROTOCOL:

1. Set Working Directory & Load Data

- **Steps:**
  - Set the working directory to where the input files are stored.
  - Load:
    - **Count matrix**: TCGA-count-data.csv
    - **Sample metadata**: TCGA-column-data.csv
- **R Code Snippet:**

```
setwd("C:/Users/keert/OneDrive/Desktop/KEER_SRA_NGS/deseq_sample/deseq_sample")
Count_data <- read.csv("TCGA-count-data.csv", header = TRUE, row.names = 1,
                                    check.names = FALSE)
Col_data   <- read.csv("TCGA-column-data.csv", header = TRUE, row.names = 1,
                                    check.names = FALSE)
```

- **Usedfor:**
  To load the **gene count matrix** and **sample metadata** into R for analysis.

2. Data Preprocessing

- **Steps:**
  - Check for missing values (NA) in the count matrix.
  - Handle missing values by imputing them using **row-wise mean** (zoo package).
  - Optionally remove genes with all zero counts.
- **R Code Snippet:**

```
library(zoo)

# Check NAs
sum(is.na(Count_data))

# Impute missing values with row mean
Count_data <- t(apply(Count_data, 1, function(x) na.aggregate(x, FUN = mean)))

# Remove genes with all zeros
Count_data <- Count_data[rowSums(Count_data) > 0, ]
```

Used for:
To clean and preprocess raw count data, ensuring DESeq2 can run without errors.

3. Prepare DESeq2 Dataset

- A **DESeqDataSet** object was created using the count matrix and sample metadata.
- The experimental design was specified as ~ condition, allowing the model to compare expression between conditions.
- The reference level for condition was set to **"normal"** so that tumor samples were compared against the normal baseline.

Used for: To organize count data and metadata into a format suitable for DESeq2 analysis.

4. Run Differential Expression Analysis

- The DESeq2 pipeline was executed to:
  - Normalize raw counts.
  - Estimate dispersion across samples.
  - Fit a negative binomial model for each gene.
- Differential expression results were extracted, providing **log2 fold changes** and **adjusted p-values (padj)**.

Used for: To statistically identify genes that are significantly differentially expressed between normal and tumor conditions.

5. Extract Significant Differentially Expressed Genes (DEGs)

- Genes were filtered based on significance thresholds:
  - **Upregulated genes:** log2 fold change > 1 and padj < 0.05.
  - **Downregulated genes:** log2 fold change < -1 and padj < 0.05.

- **All DEGs:** complete results table with log2 fold change and adjusted p-values.

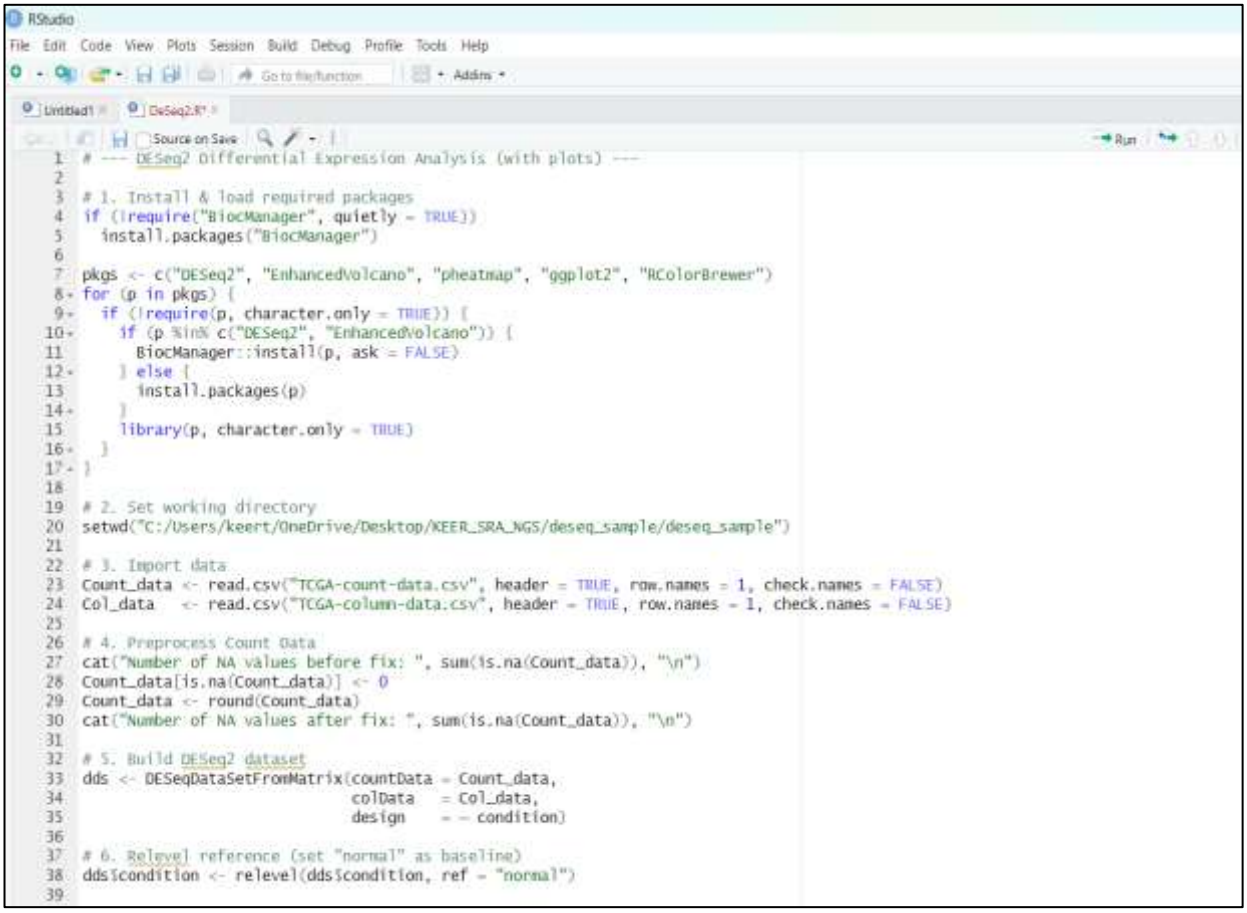
- Results were exported into separate CSV files:
  - Upregulated.csv
  - Downregulated.csv
  - All_DEGs.csv

Used for: To obtain gene lists that can be used for downstream biological interpretation and pathway analysis.

6. Volcano Plot Visualization

- A **volcano plot** was generated using the Enhanced Volcano package.
- The x-axis represented **log2 fold change** (magnitude of expression change).
- The y-axis represented **–log10(p-value)** (statistical significance).
- Genes with significant upregulation and downregulation were highlighted in different colors.
- The volcano plot was saved as both PNG and PDF files.

Used for: To visually represent significantly upregulated and downregulated genes in terms of fold change and statistical significance.



```
# --- DESeq2 Differential Expression Analysis (with plots) ---

# 1. Install & load required packages
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

pkgs <- c("DESeq2", "EnhancedVolcano", "pheatmap", "ggplot2", "RColorBrewer")
for (p in pkgs) {
  if (!require(p, character.only = TRUE)) {
    if (p %in% c("DESeq2", "EnhancedVolcano")) {
      BiocManager::install(p, ask = FALSE)
    } else {
      install.packages(p)
    }
    library(p, character.only = TRUE)
  }
}

# 2. Set working directory
setwd("C:/Users/keert/OneDrive/Desktop/KEER_SRA_NGS/deseq_sample/deseq_sample")

# 3. Import data
Count_data <- read.csv("TCGA-count-data.csv", header = TRUE, row.names = 1, check.names = FALSE)
Col_data   <- read.csv("TCGA-column-data.csv", header = TRUE, row.names = 1, check.names = FALSE)

# 4. Preprocess Count Data
cat("Number of NA values before fix: ", sum(is.na(Count_data)), "\n")
Count_data[is.na(Count_data)] <- 0
Count_data <- round(Count_data)
cat("Number of NA values after fix: ", sum(is.na(Count_data)), "\n")

# 5. Build DESeq2 dataset
dds <- DESeqDataSetFromMatrix(countData = Count_data,
                              colData   = Col_data,
                              design    = ~ condition)

# 6. Relevel reference (set "normal" as baseline)
dds$condition <- relevel(dds$condition, ref = "normal")
```

RStudio

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```r
# 7. Run DESeq2
dds <- DESeq(dds)
res <- results(dds)
summary(res)

# 8. Extract significant DEGs
resSigUp   <- subset(res, padj < 0.05 & log2FoldChange > 1)
resSigDown <- subset(res, padj < 0.05 & log2FoldChange < -1)

write.csv(resSigUp,   "Upregulated.csv")
write.csv(resSigDown, "Downregulated.csv")
write.csv(res,        "All_DEGs.csv")

# ==============================
# Visualization Section
# ==============================

# ---- A. Volcano Plot ----
vol_plot <- EnhancedVolcano(res,
                            lab = rownames(res),
                            x = 'log2FoldChange',
                            y = 'pvalue',
                            pCutoff = 0.05,
                            FCcutoff = 1,
                            xlim = c(-8, 8),
                            ylim = c(0, 12),
                            col = c("blue3", "black", "black", "red3"),
                            colAlpha = 0.9,
                            pointSize = 2.5,
                            labSize = 3.5,
                            legendPosition = "bottom",
                            legendLabSize = 12,
                            legendIconSize = 4.0,
                            gridlines.major = FALSE,
                            gridlines.minor = FALSE)

print(vol_plot)  # Show in RStudio
ggsave("volcano.png", plot = vol_plot, width = 10, height = 8, dpi = 300)
```

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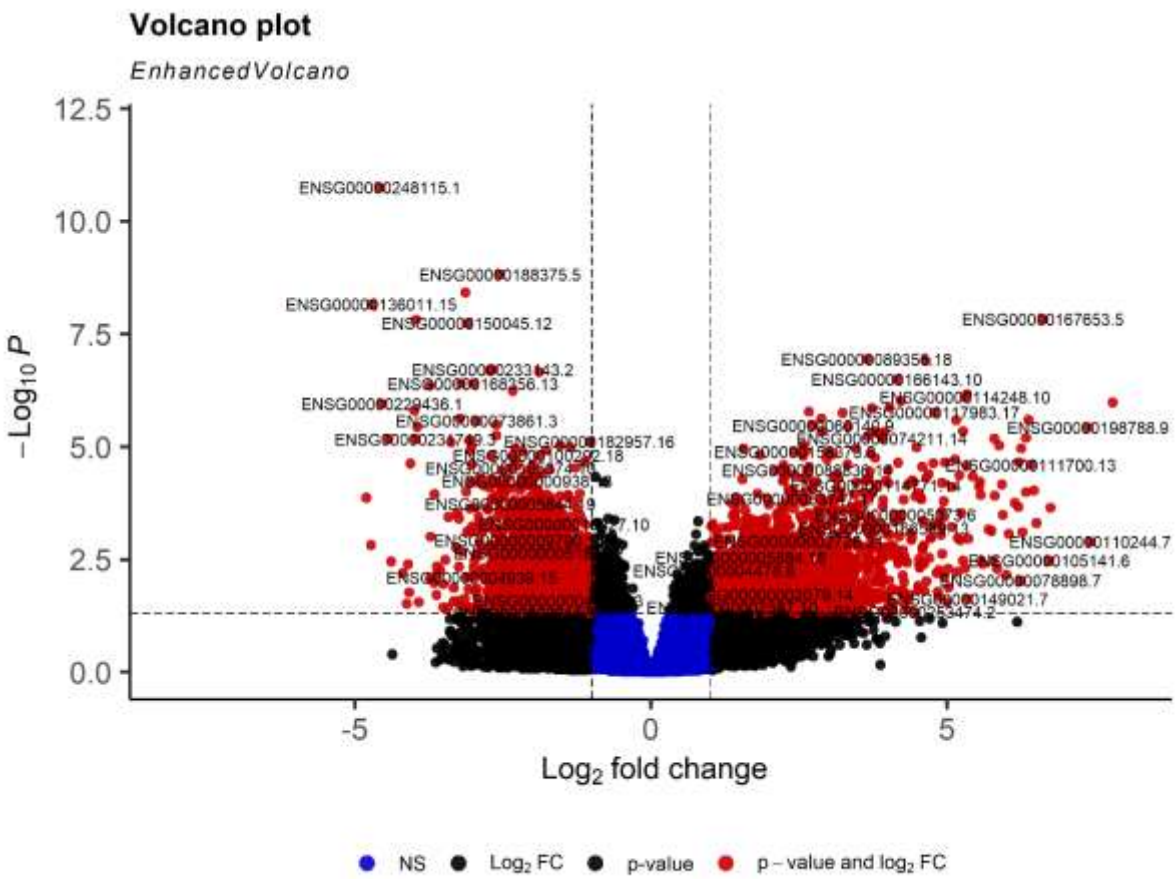
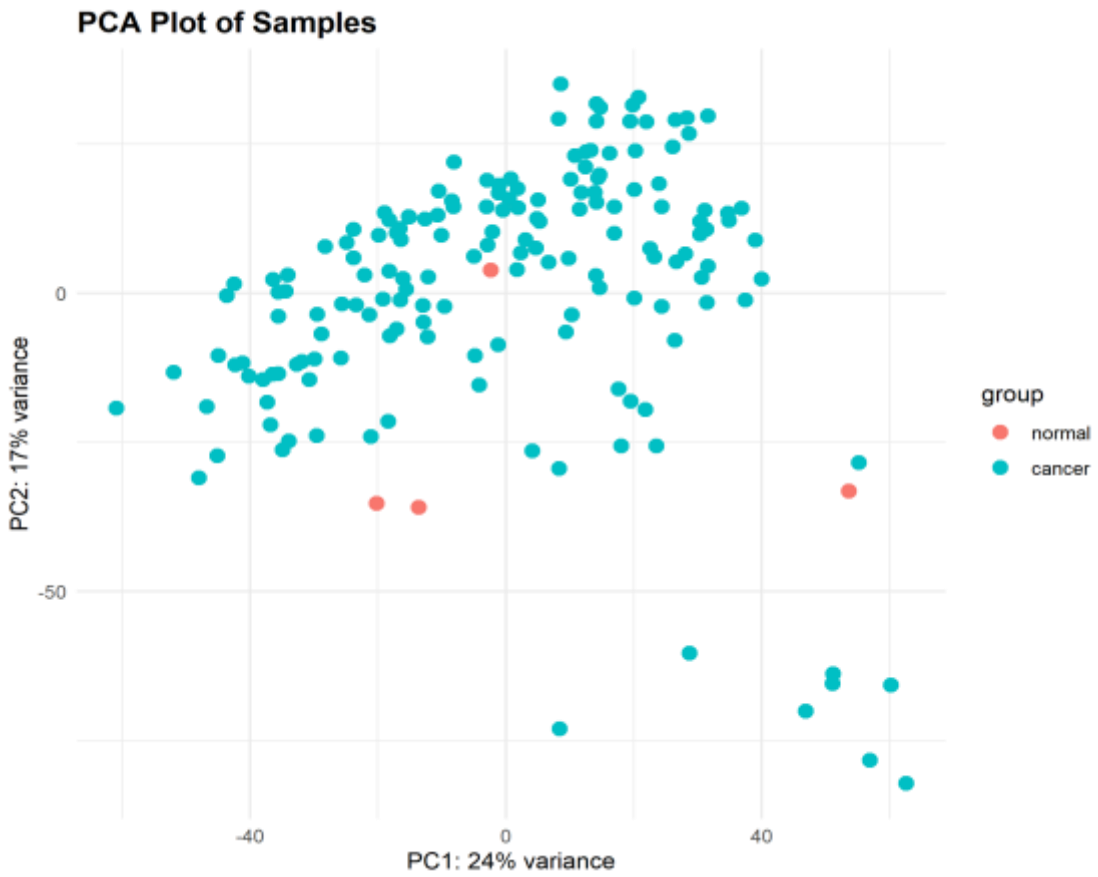
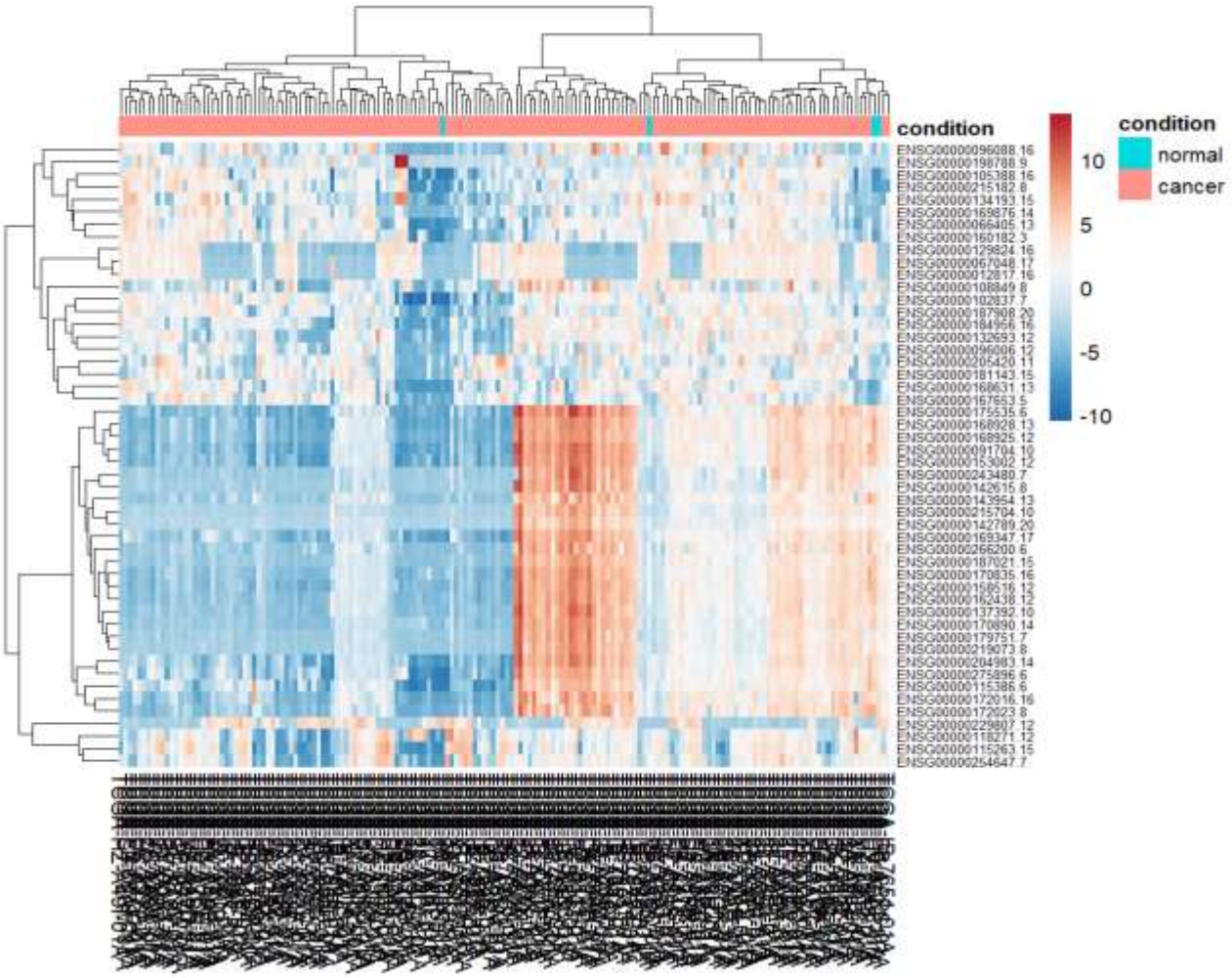
```r
                            legendLabSize = 12,
                            legendIconSize = 4.0,
                            gridlines.major = FALSE,
                            gridlines.minor = FALSE)

print(vol_plot)  # Show in RStudio
ggsave("volcano.png", plot = vol_plot, width = 10, height = 8, dpi = 300)

# ---- B. PCA Plot ----
vsd <- vst(dds, blind = FALSE)

pca_plot <- plotPCA(vsd, intgroup = "condition") +
  ggtitle("PCA Plot of Samples") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "right"
  )

print(pca_plot)  # Show in RStudio
ggsave("PCA_plot.png", plot = pca_plot, width = 8, height = 6, dpi = 300)

# ---- C. Heatmap of Top 50 Variable Genes ----
topVarGenes <- head(order(rowVars(assay(vsd)), decreasing = TRUE), 50)
heatmap_data <- assay(vsd)[topVarGenes, ]
heatmap_data <- heatmap_data - rowMeans(heatmap_data)

hm <- pheatmap(heatmap_data,
               annotation_col = as.data.frame(colData(vsd)["condition"]),
               show_rownames = TRUE,
               fontsize_row = 6,
               clustering_distance_rows = "euclidean",
               clustering_distance_cols = "euclidean",
               color = colorRampPalette(rev(brewer.pal(9, "RdBu")))(255))

# Save heatmap to PNG
png("Heatmap_top50.png", width = 1200, height = 1000, res = 150)
print(hm)   # Force render
dev.off()
```


Volcano plot
EnhancedVolcano
12.5
10.0
7.5
5.0
2.5
0.0
−Log10 P
-5
0
5
Log2 fold change
NS
Log2 FC
p-value
p – value and log2 FC
total = 60660 variables


PCA Plot of Samples
0
-50
PC2: 17% variance
-40
0
40
PC1: 24% variance
group
normal
cancer


condition
condition
normal
cancer
10
5
0
-5
-10